What Causes Turnover? Evidence from an Industrial Park in Ethiopia*

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Abstract

We study the causes of high turnover rates in a flagship industrial park in Ethiopia with detailed administrative records for 58,530 applicants during 2018–23. (1) Leveraging exogenous price shocks, we find that better outside options reduced the likelihood that applicants take any jobs in the industrial park but did not induce current workers to separate. A simple job search model can explain the effect of price shocks on applicants. (2) Using productivity data from individual firms, we find evidence that workers with lower productivity were more likely to separate. A simple model with workers learning about their productivity on the job can partially explain the correlation. The findings suggest that high turnover is not necessarily an indicator of "bad jobs", but rather a depiction of a competitive local labor market with many young, inexperienced workers.

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1 Introduction

Labor markets in low-income countries feature frequent flows between jobs (Donovan et al., 2023). Workers often transition between marginal jobs such as self employment and informal, low-earning wage work. These labor market flows do not result in meaningful wage growth, suggesting high income volatility in these labor markets.

In response, many low-income countries are actively pushing to transition away from subsistence-agriculture to formal wage work that can offer a stable income flow. Many industrial policies provide amenities and subsidies to attract large-scale, often multinational, manufacturing firms to start production in the country. Although these manufacturing firms generally offer comparatively good formal job opportunities, turnover rates are still high (Blattman and Dercon, 2018). Some policy debates use the high turnover rates as evidence against such industrial policies, as workers choose to work elsewhere instead of staying in these "bad" manufacturing jobs. However, we have little evidence of what causes workers to quit these jobs, and different causes may have different implications on the nature of these new manufacturing jobs.

This paper focuses on Hawassa Industrial Park (HIP), a flagship project in Ethiopia that received 58,530 applicants between 2018–23 and uniquely suited to study the causes of high turnover rates. We discuss two main hypotheses. First, a worker may quit because she finds a better outside option. We leverage exogenous price shocks in coffee, a major cash crop in the region, to examine the effect of outside options on turnover. Second, a worker may quit because she updates negatively on the match quality. We collect productivity data from firms in the HIP to show that there is a negative correlation between individual productivity and separation, from which we infer on-the-job learning. Despite high turnover rates, Our findings suggest that it is not necessarily an indicator of "bad jobs" in the context of HIP.

HIP is the largest employer of garment industry in the Sidama region. One unique feature of HIP is the centralized hiring system. In 2018, shortly after the HIP was established, the Ethiopian government set up an office next to the HIP to facilitate the hiring for entry-level workers. All applicants were required to register in the hiring office and would wait for an assignment to one of the firms in HIP. We obtained administrative records from the hiring office between 2018–23, with details on each worker's date of registration, origins, dates of

all assignments and terminations if any. Most applicants are female (89%), young (average age 25), with no more than 10th grade education (88%).

We define two types of turnover in HIP. First, applicants might decline an assignment and never work in HIP. This type of turnover is similar to rejecting a job offer in a job search model. In other settings where large manufacturing firms conduct hiring by themselves, entry-level job seekers are usually hired on the spot and wait for further assignments, and many would quit on the first day of work (see Blattman and Dercon (2018) for an example), which potentially reflects this type of turnover. In HIP, we observe that 34% of all applicants never took up a job, suggesting that a large proportion of them might have declined the first assignment. This is unlikely to be driven only by potential lack of jobs: on average, we observe 66 applicants per day, including those who re-entered the pool, and 53 hires per day.

Second, among the hires, some separated after staying on the job for a period of time. This type of turnover is the conventional definition of separation. Of all applicants, 43% accepted a job assignment and separated before the end of the study period. Of these workers, 42% never showed up in the record again. Given that HIP is the largest recruiter of garment industry in Sidama region, this is a strong indicator that workers switched to a different industry after separation. We thus refer to it as industry-specific separation. For the rest of workers who separated from the previous job but accepted another assignment later in HIP, we refer to it as firm-specific separation.

We now examine the first hypothesis that turnover is driven by better outside options. We first use Ethiopian Socio-Economic Panel Survey between 2021–22 and Ethiopian Agricultural Sample Survey between 2018–19 to characterize potential outside options in nearby regions.[1] More than 40% of working-age survey respondents worked in agricultural production, 10% in non-wage self employment, and less than 10% in wage employment, suggesting that agriculture is still the main sector to absorb most employment. The median salary in the HIP sits around 50 percentile of female wage employment, 50 percentile of monthly revenue from self-employment, and 60 percentile of monthly crop sales, confirming that the HIP provides relatively good, stable income flow, especially for young, inexperienced women who never worked in private sector before.

Our empirical strategy leverages exogenous price shocks in coffee, one of the most salient

[1]They include the Southern Nations, Nationalities, and Peoples' Region, or SNNP, of which Sidama was a part but separated since 2020, and Oromia region.

cash crop in Sidama region and the largest export goods in Ethiopia. We collect global coffee prices for Arabica coffee from International Monetary Fund as well as local coffee prices collected from the Ethiopia Commodity Exchange. We first show a strong, almost one-to-one correlation between local and global coffee prices, suggesting that an exogenous increase in global coffee prices can translate into an increase in local demand for coffee production, while such price shocks are orthogonal to the conditions of local labor supply in coffee production. We further assume that not all workers have complete and timely information about coffee production and that workers from coffee-producing districts are more likely to be exposed to an increasing labor demand for coffee production. Thus, we capture exposure to coffee price shocks by interacting the log price of global coffee one month after her application date with whether a worker originally came from a coffee-producing district, defined as those where the share of coffee producing fields is above the median (10.5%) in the Agricultural Sample survey. We further control for district fixed effects, month fixed effects, demographics, and the initial selection of the applicant pool by including the interaction of the global coffee price one month before and whether a worker originally came from a coffee-producing district.

We find a significant effect on the first type of turnover, that is, whether a worker never took up a job in HIP. For applicants from coffee producing districts, one standard deviation increase in coffee prices leads to a 6.5 percentage point increase in the likelihood of never working in HIP. This effect is robust to various specification tests and falsification exercises. For workers who have already accepted a job assignment, we do not find any significant impact of coffee price shocks on separation. This suggests that once workers started working in the HIP, they did not use this job as insurance against temporary income shocks, consistent with anecdotes that many workers are indeed seeking a career in the garment industry.

We then use a simple job search model à la McCall (1970) to characterize turnover. In a competitive labor market, workers choose to accept a job when the wage exceeds the reservation wage that equals the flow income from outside options. We then estimate the key distributional parameters from the turnover data using Generalized Methods of Moments (GMM), and compare these estimates with the survey measure of distributions. The model estimates show that a simple job search model can explain well the effect of coffee price shocks on whether a worker never took a job in the HIP, although it fails to predict accurately the exact level of such turnover.[2] The model also fails to explain on-the-job separation, as

[2]The latter is a common critique of the reservation wage method because salary and income are difficult

workers who took a job in the HIP would have been much more responsive to shocks in outside options under the search hypothesis.

We now examine the second hypothesis where a worker quits because she updates negatively on the match quality. In a classic paper, Jovanovic (1979) formulated that when workers learn about their productivity on the job over time, (i) those with low-level productivity would choose to quit, and (ii) the average productivity of stayers would increase. We collected detailed personnel and productivity data from five firms in HIP to directly test these two predictions. One difficulty is that almost all firms collect productivity data only at the production line level. However, assuming firms assign a new worker quasi-randomly to a production line, we can compare the productivity for the given production line before and after the worker was assigned to it, which will reflect individual-level productivity for the new worker especially at the beginning of her tenure. Using this AKM-style decomposition technique, we impute individual-level productivity within 45-day probation period for 2,602 workers and productivity within one-year period for 3,872 workers, using the percentage of targets met as the productivity measure that is consistent across all firms.

We have two main findings. First, we observe an increase in the average productivity among stayers, consistent with the second prediction from Jovanovic (1979). Second, we observe a negative correlation between individual productivity and separation. One standard deviation increase in individual productivity imputed during probation is correlated with 3.2–4.0 percentage point decrease in separation. This is consistent with the first prediction from Jovanovic (1979). We also observe a nuance with respect to industry-specific versus firm-specific separation: While individual productivity imputed during probation is mainly correlated with industry-specific separation, individual productivity imputed within one year is mainly correlated with firm-specific separation. This suggests that workers may update their own comparative advantage early in their tenure; if workers still stay on the job one year after, they may interpret productivity signals mainly as indicators of firm-specific amenities.

We then use a simple learning model à la Jovanovic (1979) to characterize turnover. Workers receive a productivity signal in the first period and update their beliefs about the true productivity in a Bayesian way, and decide to separate if the expected true productivity plus a switching cost does not exceed their outside options. In the second period,

to measure precisely and do not reflect non-pecuniary amenities (e.g. Hall and Krueger (2012).

after the initial selection, workers make quitting decision again based on the average productivity signals received thus far. Estimating the key learning parameters using GMM, the model successfully predicts that workers place a high weight on initial productivity signal to learn about their true productivity. However, the simple model cannot explain the nuance regarding industry-level versus firm-specific separation.

Our findings thus suggest that turnover can be explained by workers searching for different jobs at the application stage and partially by learning about their true productivity on the job. Neither explanation inherently indicates that the HIP jobs are heavily disliked; if anything, our estimates from the search model suggest that HIP jobs are placed at a higher percentile in some job distributions compared to the survey measures. Instead, turnover characterizes the job distribution in the local labor market, and its reaction to exogenous shocks in outside options can be described with a standard job search model in a competitive labor market. It also depicts the nature of the young, inexperienced workforce who are still trying to identify their comparative advantages. In this sense, high turnover is beneficial for workers to target better job opportunities in the long run (Gorry, 2016), but less so for firms who bear the training cost upfront (Pallais, 2014).

Our paper first contributes to understanding the causes of high turnover rates in manufacturing industries. The early literature documented high turnover in the early stages of industrialization (Montgomery, 1989; Beckert, 2015) or among young workers in the United States (Farber, 1994, 1999). More recent work has found high turnover rates in developing countries (Groh et al., 2016; Blattman and Dercon, 2018). The understanding of causes is mostly speculative, mainly due to the lack of personnel records of individual workers and the difficulty of identifying workers' outside options. Our paper leverages the rich personnel data available in HIP as well as the significance of coffee production as a major outside option in the nearby regions, to provide identification for the causes.

Second, we contribute to the understanding of the features of labor markets in low-income countries. A growing literature documents the existence of frictions that prevent these labor markets from being perfectly competitive (see Breza and Kaur (2025) for a review).[3] Our turnover analysis suggests that although turnover can be characterized by job search in a

[3]An incomplete list of papers on search frictions: Abebe et al. (2021); Bandiera et al. (2023); Caria et al. (2024); Franklin (2018); Kelley et al. (2023); Fernando et al. (2023); Hensel et al. (2024); Wu and Wang (2026). Information frictions: Abebe et al. (2024); Abel et al. (2020); Alfonsi et al. (2023); Banerjee and Sequeira (2023); Bassi and Nansamba (2022); Beam (2016); Carranza et al. (2023).

competitive labor market, inexperienced workers learning about their productivity on the job is potentially inefficient (Pallais, 2014). High turnover induced by learning reflects such an inefficiency of the labor market, which is potentially more salient in low-income contexts where a large proportion of the workforce does not have industry experience.

Third, our findings have implications on the impact of large multinational firms on local labor markets. Recent literature shows positive labor market effects of large multinationals in higher-income contexts (Alfaro-Ureña et al., 2021; Setzler and Tintelnot, 2021). The relevant policy discussions in low-income countries, unfortunately, often rely on indirect evidence due to the lack of data. Our evidence, although still indirect, provides a more optimistic view of large multinational firms in low-income countries, that they offer comparatively good jobs for the local population and training for the inexperienced workforce. Future research is needed to understand whether these jobs provide a starting point for these young workers in pursuit of better careers in the future.

The rest of the paper is structured as follows. Section 2 describes the context and the turnover data. Section 3 discusses search hypothesis. Section 4 discusses learning hypothesis. Section 5 concludes.

2 Turnover in HIP

2.1 Hawassa Industrial Park

Ethiopia is a low-income country in east Africa, with a GDP per capita of 944 US dollars in 2021. Starting from the early 2000s, the government endeavored to attract foreign investment opportunities, predominantly in the light manufacturing sector, in hope to transition the economy from agriculture-based to industrialization. In particular, many East Asian companies, state-owned or private, have invested in many infrastructure projects and the light manufacturing sector, mainly because of the relatively low labor cost in Ethiopia and the large subsidies promised by the government.

One of the major government-led project is the Hawassa Industrial Park (HIP), a flagship project in Ethiopia and one of the largest industrial parks in sub-Saharan Africa. It is located in Sidama region, traditionally an agrarian society featuring cash crops such as coffee beans. Around 20 firms operated in the industrial park between 2018–2023, all but

one in the garment sector.[4] Within the years where we can observe turnover statistics, HIP has registered 58,530 applicants, actively making itself the major recruiter of the garment sector in southern Ethiopia (Hardy et al., 2022). There is no other competitor of similar size in the region; all other industrial parks are located at least 200km away from the region.

One unique feature of the HIP is the centralized hiring system. To support the hiring process of entry-level workers, the Ethiopian government established a hiring office next to HIP to facilitate the hiring process. Job seekers in the nearby area can directly walk in and register for a job. In addition, the government set up ten recruitment centers within 60km around HIP and recruited workers from outside of Hawassa. After registration, job seekers would be randomly assigned to one of the firms requesting for workers. This centralized hiring system essentially formed a monopsony in the labor market, as firms in HIP agreed to only use the centralized hiring system to hire entry-level workers and not poach from each other. Importantly, it is difficult for an entry-level worker to choose their employer based on better job aspects because they have to re-enter the labor pool and wait for a new draw. If a worker leaves HIP, given the dominant size of HIP in the garment industry in Sidama region, it is likely that the worker would be employed in a different industry.

The criteria for applying for a job in the HIP are very low. Anyone at least 18 years old with eighth grade education is qualified for a job in the HIP. Firms tend to prefer female workers, but they accept over 95% of the first-assigned workers. In fact, 89% of all applicants are female. The ones rejected by firms are usually the ones who used to work in the firm and were already fired by the firm once before, or because firms requested workers for specific tasks for which they have more stringent criteria.

2.2 Turnover Data

We collected administrative records from the hiring office between 2018–2023. The data include all 58,530 applicants who registered in HIP between 2018–2023. Each record includes the date of registration, basic demographics (age, gender, education), whether the worker self-applied or was sourced from the nearby recruitment centers, and the origin of the worker.

[4]Since 2021, the civil war and the termination of African Growth and Opportunity Act (AGOA) agreement in Ethiopia heavily affected the exporting industries, especially firms who predominantly exported to the United States and Europe. Nevertheless, many companies are from East and South Asia whose main exporting markets are not in the US or Europe, therefore less affected as of September 2023.

If a worker was ever allocated to or fired from a company, we observe the date of allocation or termination and which company.

We use this data to define two types of turnover in this context. First, applicants may have registered but then chose to decline a job assignment. This type of turnover is closer to rejecting a job offer in a standard job search model. It is potentially relevant to many other settings where large manufacturing firms conduct hiring themselves (e.g. Blattman and Dercon (2018)). Job seekers would be "hired" on the spot and wait for task assignments; meanwhile, they can easily leave and seek for another job before committing to the first assignment. We do not directly observe whether an applicant rejected an assignment in our data. However, 34% applicants never took up a job during the study period. This number goes up to 36% if we only include applicants who we can observe for at least one year during the study period. This suggests that a substantial share of job seekers might have declined the first job assignment.

Could it be because applicants simply did not get a job assignment? Our data suggests unlikely. On average, we observe 66 applicants per day in the hiring office, including those re-entering the labor pool, and 53 hires per day, which indicates potentially sufficient labor demand. Figure 1, Panel A shows the percentage of first-time applicants who haven't taken a job within a certain period. Almost 40% first-time applicants took a job within one day of application, but this number quickly plateaus after the first week. Suppose a worker shows up five days a week with a 40% likelihood to get a job assignment per day, the probability of getting a job assignment within one week is 92%. We therefore believe that the 34% of applicants never taking up a job mainly reflects applicants' rejection of a job assignment.

Second, among job seekers who accepted the first job assignment, some separated after working on the job for some time. This is the conventional definition of turnover (Farber, 1994). Figure 1, Panel B shows the percentage of workers separating from the first job within one day to within one year. 4.6% workers separated within one day, 14.5% within one month, 32.3% within one year. These two statistics are less than half the magnitudes as in Blattman and Dercon (2018). Overall, 43% workers separated from the first job within the observation window.

Among the workers who separated, some eventually came for another job assignment. We refer to it as firm-specific separation, as these workers may not enjoy the previous firm-

specific match but still value another job in a different HIP firm. The rest is referred to as industry-specific separation, as these workers never returned for another job assignment and likely employed in a different industry elsewhere. Figure B1 shows the percentage of industry-specific separation within one day to within one year. Only 0.9% workers separated permanently within one day, 1.7% within one month, 8.0% within one year. Overall, 18% workers separated from the first job within the observation window and never returned.

In summary, of the 58,530 applicants, we observe that 34% never took up a job assignment, 43% separated from the first job, among which 42% (=18%/43%) never returned to HIP. The two types of turnover statistics add up to 77%, close to the statistics observed in Blattman and Dercon (2018), potentially with different causes and implications.

2.3 Can Labor Demand Shocks Explain Turnover?

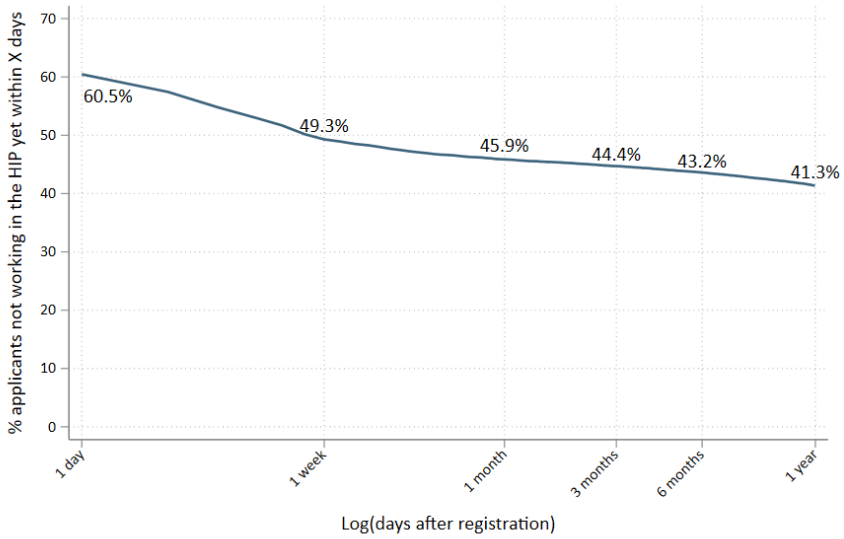
Before we examine the two supply-side hypotheses of turnover, we briefly discuss whether labor demand shocks potentially cause turnover. Although anecdotally, firms rarely fired workers after the probation period before Covid-19, there are more such occasions after 2020. Figure B2, Panel A shows the monthly trend of separation during the study window. We see a spike shortly after the lockdown in March 2020, and a few more spikes during Ethiopian civil war, especially around the end of 2021 when Ethiopia was no longer included in the African Growth and Opportunity Act (AGOA). The volatility in firm production could potentially lower the labor demand and cause involuntary turnover.

We choose not to focus on involuntary turnover for three reasons. First, under certain hypotheses, involuntary turnover has the same efficiency implications as voluntary turnover. Jovanovic (1979) famously pointed out that if workers quit because they realize a bad match component, such a signal would also be observable to firms. Whether workers quit or are fired by the firm due to a bad match is essentially equivalent. Another example is that firms may not open many vacancies when they expect workers to not take the job. This, however, is an equilibrium behavior in response to workers' voluntary turnover.

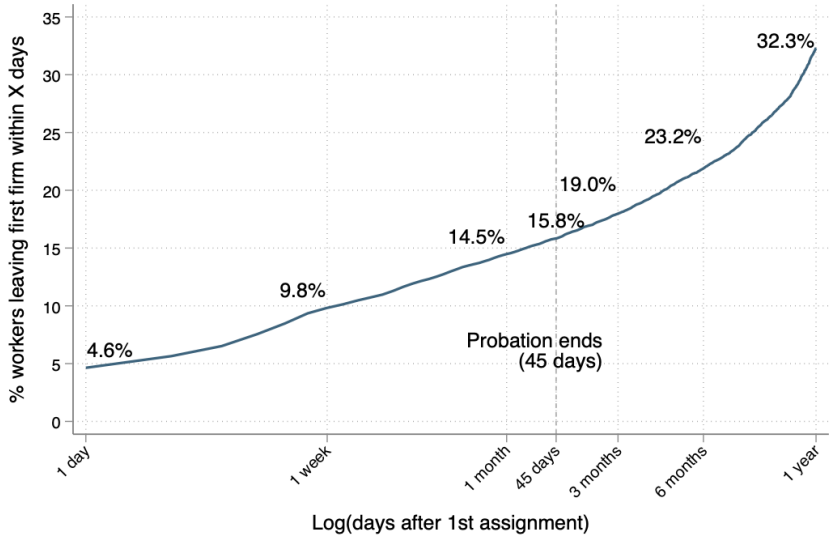
Second, there is no consistent data on termination reasons. We only observe termination reasons for applicants before 2020. Even so, sometimes a worker is "fired because of absences". It is thus difficult to identify involuntary turnover as this may be a reaction to issues from the labor supply.

Figure 1: Turnover Patterns in the Industrial Park

Panel A. Percentage of Workers Not Taking A Job Yet



Panel B. Percentage of Workers Separating From the Job



Notes: This figure shows the distribution of the turnover in the industrial park between August 2018 and September 2023. We show the key statistics of turnover within one day, one week, one month, 45 days when the probation period ended, three months, six months, and one year.

Third, mass layoff events are only occasional during the study period. Figure B2, Panel B shows the distribution of the number of daily separations. The median is eight separations a day across all firms in HIP. In one of the robustness checks, we will drop the observations during the mass layoff episodes, defined as above 95 percentile in Figure B2, Panel B, and results remain largely unaffected.

3 Search Hypothesis

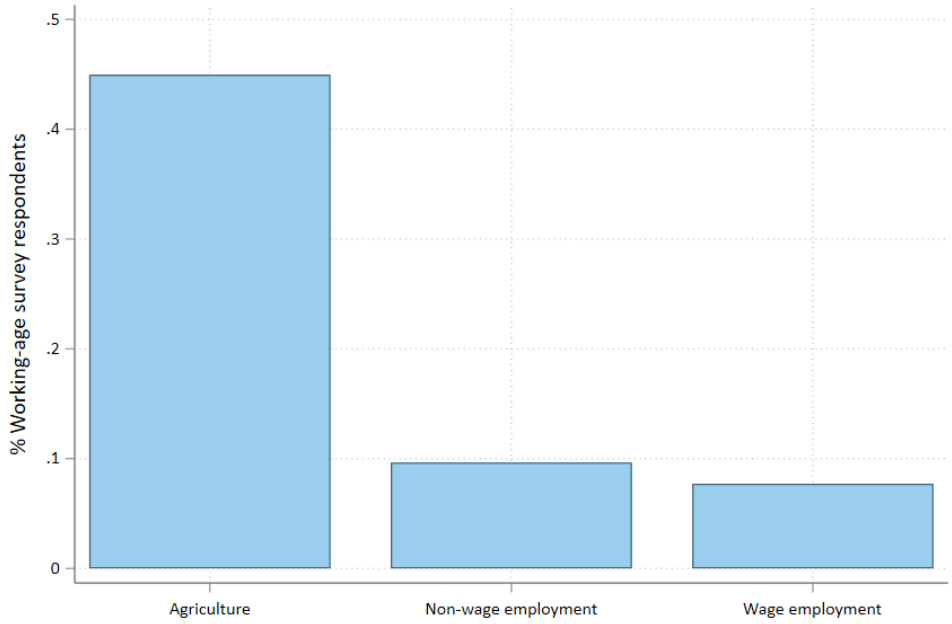
Is turnover driven by workers seeking for better job opportunities? We first characterize outside options in nearby regions with existing data such Ethiopian Socio-Economic Panel Survey (ESS, conducted by World Bank in 2021-22) and the Agricultural Sample Survey (AgSS, conducted by Central Statistical Agency during 2018–19). AgSS sampled around 27,000 farming households and 240,000 fields from 579 local districts ("woredas") across the country. We then describe our empirical strategy using global coffee prices to proxy exogenous demand shocks outside of HIP. We use a simple job search model to discuss how well search hypothesis can explain turnover.

3.1 Outside Options in Nearby Regions

We use the 3,984 ESS survey respondents aged between 15 and 64 from Southern Nations, Nationalities, and Peoples' Region (SNNPR) and Oromia for the descriptive statistics. Figure 2 shows the percentage of respondents engaged in different types of employment in the last seven days. The majority of respondents (45%) reported income-generating activities in agriculture, some in non-wage employment including self-employment (10%). Only 7.7% respondents reported any wage employment in the last seven days, 10% in the last year. Figure B3 further shows the types of employment for male and female, separately for urban and rural respondents. In general, male respondents had higher labor force participation. More urban respondents were engaged in wage employment (22% for male, 12% for female). Very few rural respondents, however, reported any wage employment (3.9% for male, 1.8% for female). This suggests that the majority of outside options are still in agriculture production, especially for the first-time job seekers from rural area.

We further show the distribution of occupations and industries for wage employment in

Figure 2: Distribution of Types of Employment in Nearby Regions



Notes: We use Ethiopian Socio-Economic Panel Survey 2021–22 to describe the distribution of types of work in SNNPR and Oromia regions. Only individuals aged above 15 and below 64 are included.

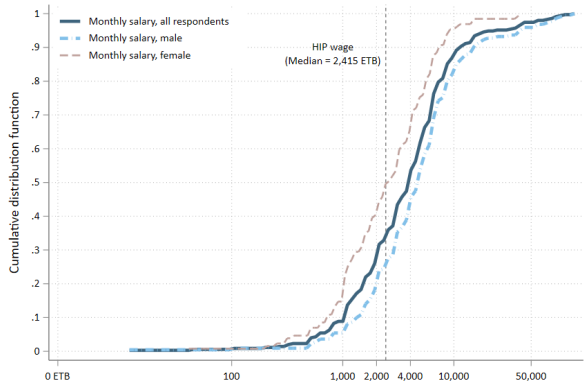
Figure B4. Notably, very few respondents worked in the manufacturing industry (5.1%) or related occupations (4.3% as plant and machine operators and assemblers); the patterns are similar for male and female workers separately. The majority of respondents work in occupations such as service workers, professionals or technicians, and office clerks, and in industries such as financial intermediation, real estate, and education. This further confirms that HIP is uniquely one the most influential recruiters in manufacturing sector, not just in garment industry.

We plot the distribution of the monthly salary of wage employment in Figure 3, Panel A. The median salary of wage employment is 3,934 ETB; male workers have higher median salary compared to female (4,500 ETB vs. 2,500 ETB). A median worker in HIP in 2022 can earn 2,415 ETB every month, roughly at 26 percentile for male workers and at the median for female workers. Some workers may choose to engage in self employment instead of salaried work. We plot the distribution of the monthly revenue, monthly profit, and monthly profit per working adult in the household for self employment in Figure 3, Panel B. Although there might be some successful cases in the far right tail of the distribution, more than

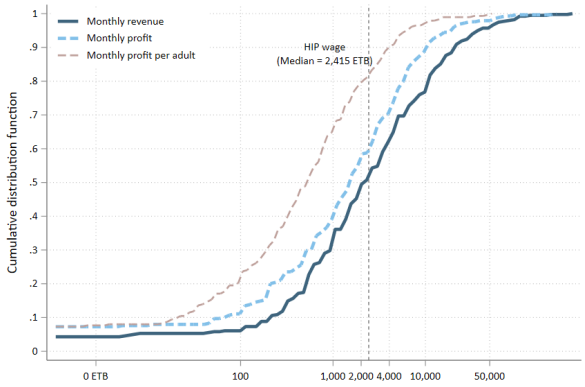
50% of self-employed respondents report a monthly revenue less than the median salary in HIP. Considering other operational costs and potential profit sharing with other household members, self employment becomes even less attractive. This is consistent with the fact that most self employment is engaged in low-capital activities, such as street vending and home baking.

Figure 3: Distribution of Monthly Income in Non-Agricultural Sectors

Panel A. Wage Employment



Panel B. Self Employment



Notes: We use Ethiopian Socio-Economic Panel Survey 2021–22 to describe the distribution of types of work in SNNPR and Oromia regions. Only individuals aged above 15 and below 64 are included. For wage employment, we use monthly salary for individuals aged between 15–64 in SNNP and Oromia. For self employment, we use monthly revenue, monthly profit (net of all operational costs), and monthly profit divided by the number of working adults in the household (aged between 15 and 64).

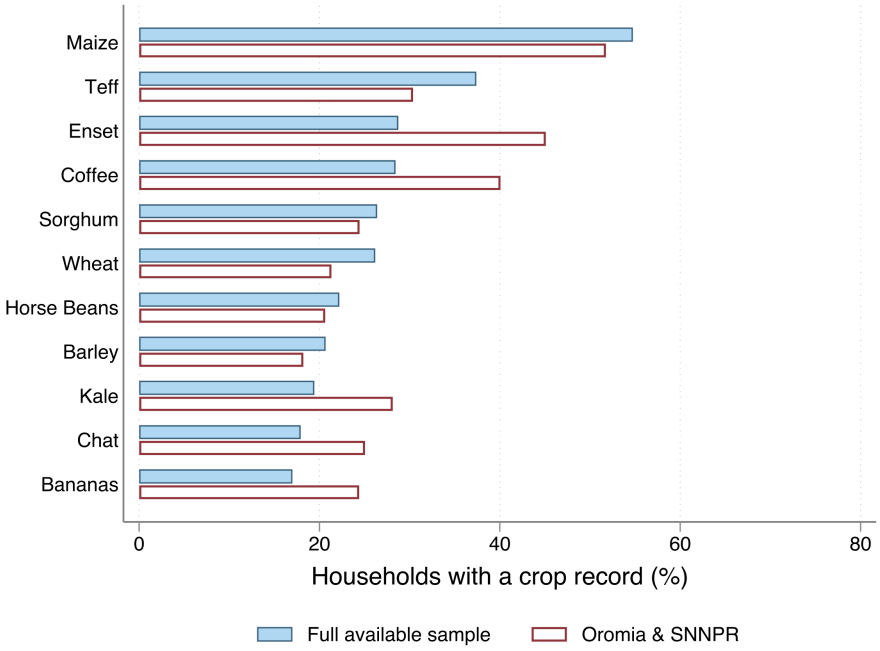
For agricultural sector, we use two data sources to impute the outside options. First, AgSS provides information about crop production from a large representative sample at the district level, but with no price information. Figure 4 shows the most common crops produced in the national sample as well as in Oromia and SNNPR. Most rural households grow staple crops such as maize, teff, and enset (or false bananas); many also grow cash crops such as coffee and chat (or qat, a recreational stimulant popular in East Africa).

Second, ESS provides detailed information about crop sales, but from a smaller sample that covers SNNPR and Oromia. For each type of crop, ESS records the total quantity harvested in the last 12 months, the total quantity for sale, and the value of all sales, from which we can impute the price of crops at the point of transaction. We then calculate the average price for each crop and impute the value of sales if all harvested quantities were for sale.[5] Figure 5 shows the distribution of the imputed agricultural income. With the

[5]Another data source from World Bank, Monthly Food Price Estimates by Product and Market, doc-

caveat that price information is often noisy from survey data, if we assume rural households perfectly smooth out income throughout the year, then about 60% of rural households have an average monthly income below HIP median wage.

Figure 4: Common Crop Production in Agricultural Sector



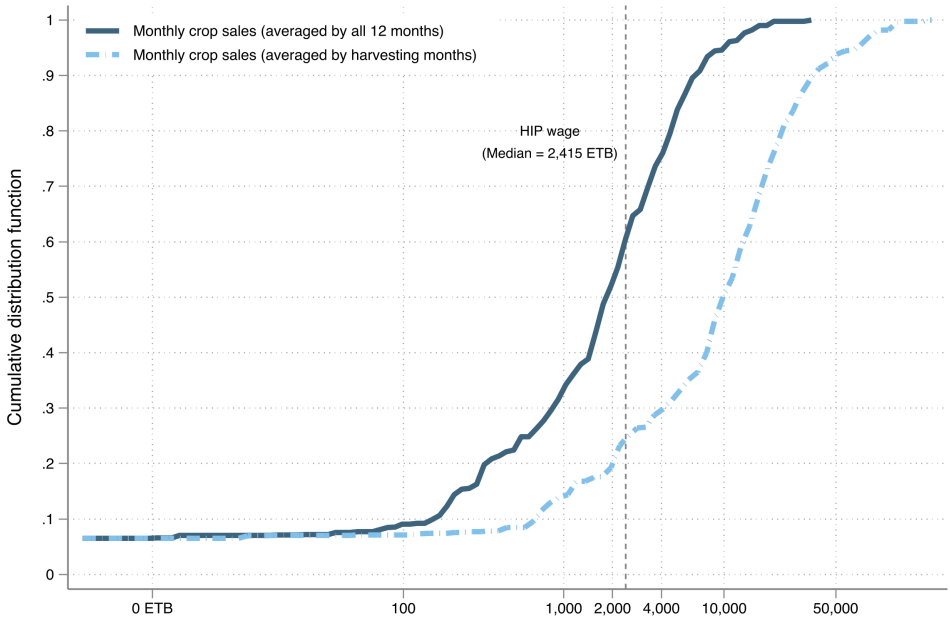
Notes: We use 2018/19 Ethiopian Agricultural Sample Survey (AgSS) for the crop-field and household data. All estimates use survey sampling weights and shows the percentage of households with at least one record of each crop.

However, this comparison does not account for the seasonality in crop production. ESS also documents the harvesting months for each crop. On average, most harvesting seasons last only 1-3 months; the existing literature has also documented that rural households tend to consume more during harvesting seasons and less in lean seasons (Fink et al., 2020). In Figure 5, we plot another income distribution, assuming that rural households only smooth income during harvesting seasons. Using this as the benchmark, about 70% rural households would have an average monthly income above HIP median wage. These descriptive statistics illustrate that agricultural production can indeed act as a competitive outside option for HIP workers, especially during harvesting seasons. Figure B5 replicates the distribution

uments prices at the market level, but it only has consistent data for staples such as maize or teff, not including cash crop such as coffee.

with quantity data from AgSS and imputed prices calculated from ESS, assuming crop production in 2018/19 is similar to that in 2021/22. Under this imputation, agricultural production becomes even more attractive.

Figure 5: Distribution of Imputed Monthly Income in Agricultural Sector



Notes: We use Ethiopian Socio-Economic Panel Survey 2021–22 to describe the distribution of imputed agricultural income in SNNPR and Oromia regions. For each crop production, we multiply the total quantity harvested in the last 12 months for each field with the average price across all households in the sample. The blue solid line plots the CDF of imputed income divided by 12. The light blue, dashed line plots the CDF of imputed income divided by the average number of harvesting months for each crop.

To summarize, our description of outside options in the nearby regions of HIP suggests that agriculture remains the biggest sector to absorb the workforce, especially for those with rural origins and potentially less advantaged in job search in the urban sector. Although agricultural production may generate less income than working in HIP or other wage employment for the majority of rural population, it can potentially provide attractive outside options during harvesting seasons. The descriptive evidence above, however, is suggestive at best, with the standard caveat of noisy measurement of agricultural production. We will discuss our causal identification strategy in the following subsections by leveraging the most prominent cash crops in Sidama region — coffee.

3.2 Leveraging Coffee Price Shocks for Identification

Recall from Figure 4 that coffee is one of the most common crops in Oromia and SNNPR. Unlike staples such as maize and teff, more than 70% of the coffee production is for sale. Figure B6 further ranks crops by the imputed gross output values and confirms the economic importance of coffee in agricultural production. More importantly, coffee is the most important export good in Ethiopia, yet Ethiopian coffee exports only account for roughly 6–7% of the global market (United States Department of Agriculture). Our empirical strategy thus leverages global price shocks as an exogenous demand shifter for coffee production in Ethiopia, eventually creating more lucrative but temporary outside options for job seekers.

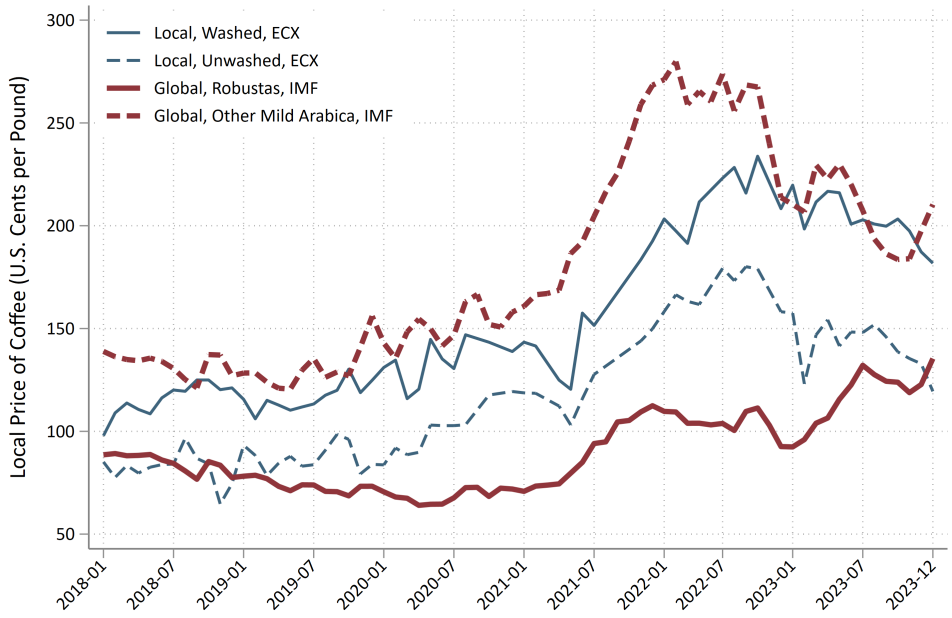
We first show that positive global price shocks are potentially correlated with higher labor demand for coffee production. The survey-based price data does not have high-frequency observations for coffee prices at the market level. Instead, we collect daily local coffee prices from the Ethiopia Commodity Exchange (ECX), which documents daily transaction prices for local washed and unwashed coffee for ECX-traded coffee contracts at the warehouse level, with the caveat that these warehouse transaction prices can be different from the retail prices at the market level. We also collect global prices on Robusta coffee and other mild Arabica coffee from the International Monetary Fund. Figure 6 presents monthly price trend of each type of coffee and shows that local prices largely comove with global prices. Table B1 shows the correlation between log prices. In general, the global price shocks translate to the local price shocks almost one-to-one, suggesting that global price shocks in coffee are likely to increase local demand for coffee.[6] The global prices of other mild Arabica coffee have higher explanatory power for both local washed and unwashed coffee, consistent with the fact that Ethiopia mainly produces Arabica coffee. We will use global prices for other mild Arabica coffee for most specifications.

We further assume that it is costly for job seekers to search for opportunities in coffee production in nearby regions. Transportation costs are already high in urban areas and prevent workers from seeking job opportunities in general (Franklin, 2018; Abebe et al., 2024; Wu and Wang, 2026). It is potentially even more difficult to obtain timely information

[6]One possibility is that, when global prices increase, intermediaries temporarily increase the trading volume with local suppliers due to higher profit margin. This can eventually transfer the global demand shock to a local demand shock. A competitive market for intermediaries may explain why global price shocks can translate to local price shocks almost one-to-one.

about temporary increases in coffee production unless workers come from a coffee producing district. Figure 7 presents the geographical distribution of workers' origins as well as the share of fields that produce coffee. Figure B7 shows the number of workers per district observed in the grading center data. There is no clear geographical pattern of coffee production, and its distribution does not coincide with the distribution of workers' origins. This provides sufficient geographical variations to examine the effect of exogenous coffee price shocks.

Figure 6: Time Trend of Coffee Prices



Notes: This figure shows coffee price trend between 2018 and 2023 (unit: US cents per pound). The local coffee prices are obtained from Ethiopia Commodity Exchange. The global coffee prices are obtained from International Monetary Fund.

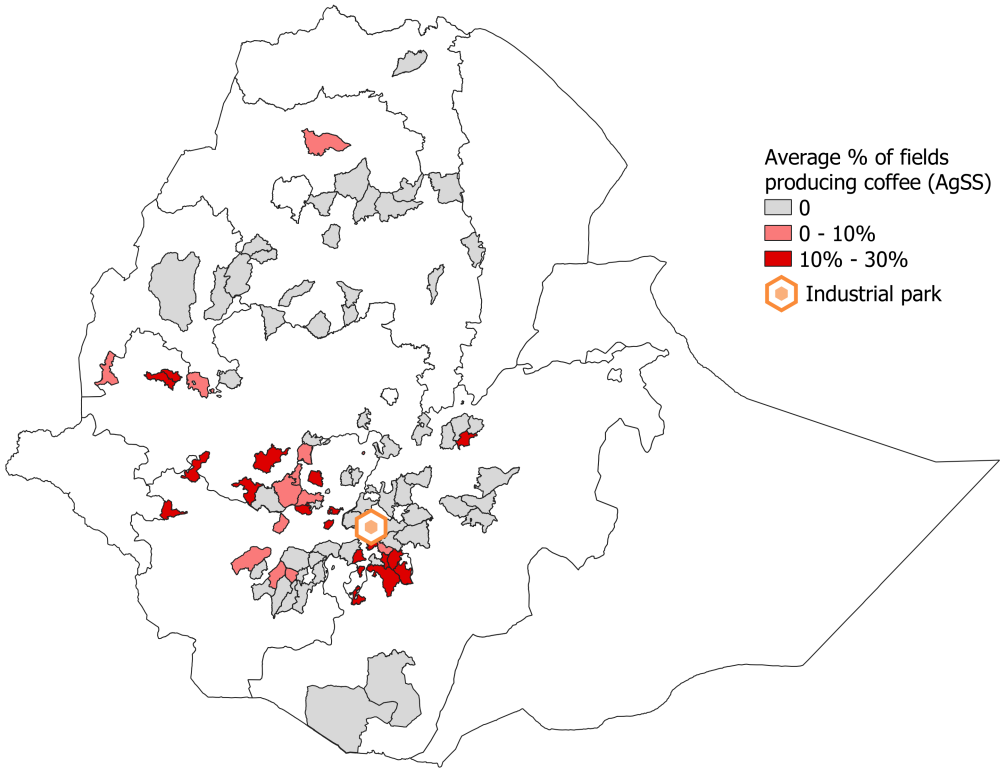
The following is our main specification to causally estimate the effect of coffee price shocks on turnover:

$$Y_{ijt} = \alpha_j + \alpha_t + \beta C_j \cdot \log P_{t+1} + \gamma C_j \cdot \log P_{t-1} + \delta X_{ijt} + \epsilon_i \tag{1}$$

Y_{ijt} is the turnover outcome for worker i originating from district j who applied for a job in the HIP in time t (year-month). C_j indicates whether district j is a coffee-producing district, defined as origins with above-median share of fields producing coffee (the median is 10.5%). P_{t+1} is the global price of Arabica coffee one month after the worker applied for

a job in the HIP. The interaction of C_j and $\log P_{t+1}$ thus captures the upcoming exogenous demand shock for coffee production that will mainly affect workers from coffee-producing districts. After controlling for time fixed effects α_t and district fixed effects α_j, β can be interpreted as the magnitude change in outcome Y_{ijt} caused by a 100 percentage change in global coffee prices among workers from coffee-producing districts. We further control for the interaction of C_j and the log global price of Arabica coffee one month before $\log P_{t-1}$ to capture potential sample selection induced by coffee prices prior to workers' joining. X_{ijt} captures other observable worker's characteristics, including gender, education, and age.

Figure 7: Distribution of Coffee Producing Districts



Notes: This figure shows the distribution of coffee producing districts. A coffee producing district is defined by whether the share of coffee-growing fields in the Agricultural Sample Survey is above median. We only show the districts from which at least one worker in the industrial park was originated.

3.3 Results

Table 1 presents the main estimates from Equation 1. We observe a significant effect in Column 1: For applicants from coffee producing districts, a 100% increase in global coffee

prices leads to 30.3 pp increase in the likelihood of an applicant never taking a job in HIP. If we look at one standard deviation increase in coffee prices (mean \$89.9, sd \$19.3), this translates into a 6.5 pp increase. Among applicants who took a job in HIP, coffee price shocks do no significantly increase the likelihood of applicant separating from the job. We further break down separation into two types: Industry-specific (quitting and never returning to HIP, most likely engaged in a different industry than garment) or firm-specific (quitting and returning to a different firm in the HIP during the observational window). We do not observe a significant effect of price shocks on either type of turnover. For a subset of applicants, we observe them multiple times in the turnover records. After controlling for applicant fixed effects in Column 5, we do not see a significant impact on separation, suggesting that within the selected sample of applicants who took a job, the effect on separation cannot be explained by individual characteristics.

To examine whether global price shocks affect turnover through local demand for coffee production, Table B2 replicates the main result in Table 1, Column 1, by replacing global price shocks with local price shocks. Column 1 shows that price increases in local washed coffee would increase the likelihood of a worker never taking a job, although insignificantly; this may reflect the endogenous supply shock in the nearby regions captured in the local prices. Column 2 shows a zero effect of price shocks of local unwashed coffee on turnover. These results, although not causal, suggest that workers react more to the price shocks in local washed coffee, potentially more sought after by buyers with higher profit margins. Column 3 instruments prices of local washed coffee with global prices of Arabica coffee and recovers the significant effect on whether a worker never took a job in the HIP. Column 4 uses global prices of Robusta coffee as the instrument, which is less affected by local supply shocks in principle because Sidama region mainly produces Arabica coffee. We still find a significantly impact on turnover, with the magnitude closer to the OLS estimate in Column 1. Together, our evidence suggests that global price shocks increase the likelihood of workers not taking any jobs in the HIP through increased demand for local washed coffee, which potentially generates more lucrative outside options for job seekers temporarily.

One may expect that such effects on turnover are more salient during the harvesting season. Table B3 breaks down the main effect by whether an applicant applied one month before the harvesting season (October to December). Interestingly, we do not observe any heterogeneity regarding harvesting season. There are three potential interpretations. First,

there can still be coffee-related job opportunities during the lean season as long as some rural households have the saving technology to preserve enough coffee beans for sale. In this case, lean season may actually provide more lucrative outside options because the profit margin can be higher due to the lack of local supply (Burke et al., 2019). Second, some firms in the HIP also developed strategies to discourage workers' temporary leaves during the harvesting season, for example, an additional attendance bonus. As a result, coffee production may not seem particularly more attractive during the harvesting season.

Table 1: Causal Effect of Coffee Price Shocks on Turnover

	(1)	(2)	(3)	(4)	(5)
	Never worked	Separated			
	in HIP	All	Not returned	Returned	All
$C_j \times \log P_{t+1}$	0.303***	0.0815	0.0224	0.0590	-0.0356
	(0.0968)	(0.128)	(0.0934)	(0.0881)	(0.229)
	[0.00281]	[0.528]	[0.811]	[0.505]	[0.877]
Observations	14,873	9,533	9,533	9,533	2,081
R-squared	0.404	0.137	0.148	0.075	0.690
Control mean	0.424	0.201	0.0816	0.119	0.201

Notes: $C_j = 1$ for origins at or above the applicant-sample median coffee exposure. Coffee exposure is the unweighted share of fields reporting coffee in Agricultural Sample Survey; the median is 10% among districts where at least one applicant in HIP was originated. Urban Sidama/Hawassa city is excluded; rural Hawassa/Awasa Zuria remains included. Month t is initial date of registration in Column 1 and the first day of work in Columns 2–5. P_{t+1} is the global price of other mild Arabica coffee at $t+1$. Dependent variable: Column 1, whether a worker never took up a job in the HIP and left no hiring records in the HIP administrative data. Column 2, whether a worker quit their first job. Column 3, whether a worker quit their first job and never returned to the HIP within the observational window. Column 4, whether a worker quit their first job and returned to the HIP within the observational window. Column 5, whether a worker quit their current job. All columns include gender, education, age, year-month fixed effects, origin fixed effects, two-way clustering by origin and year-month; we further control for $C_j \times \log P_{t-1}$ to control for potential selection induced by coffee prices at $t-1$. Column 5 replaces origin fixed effects with worker fixed effects and clusters at the worker level. Low-coffee reference means use the pooled eligible applicant/episode samples before regression exclusions, including in Column 5. Standard errors in parentheses; p-values in brackets. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Third, job seekers during the lean season may anticipate an increase in coffee prices in the upcoming harvesting season. Table B4 replicates the results with average coffee prices between 2–12 months after application. For workers from coffee-producing districts, a 100% increase in global coffee prices in later months leads to a 14.7 pp. increase in the likelihood of never working in the HIP, although it is precisely estimated (p-value 0.139) and has a lower magnitude. This suggests that the lack of heterogeneity regarding harvesting season can

be, at best, partially explained by the anticipation effect. Table B4 also does not show any significant impact on any separation outcomes with later price shocks. Our evidence thus suggests that although outside option hypothesis may explain turnover at the application stage, it fails to explain separation once a worker joins a firm.

We conduct multiple robustness tests of the main result in Table 1, Column 1. First, Table B5 conducts multiple specification tests of the main result. Column 2 and 3 use logit and probit regressions and find similar results. Column 4 uses price level instead of log of price. The result is significant despite the estimated magnitude halved — for one standard deviation increase in price, Column 4 suggests a 3.2 pp increase in the likelihood of a worker never taking a job. Column 5 uses a 0-1 indicator of price increase ($P_{t+1} > P_{t-1}$) as the independent variable and shows a very similar magnitude of effect (5.7 pp). Column 6 uses global price of Robusta coffee instead of Arabica coffee and shows a positive effect with similar magnitude. Column 7 drops mass layoff episodes and finds a similar although less significant effect. Column 8 controls for district-month fixed effects concerning district-specific seasonality; the result remains unchanged. Our main result is thus robust to various specifications. Second, we examine whether our specification captures price shocks of other crops instead of coffee. Table B6 replaces coffee price shocks with prices of other crops (maize, teff, wheat, sorghum), which we obtain from the World Food Programme market-monitor quotations.[7] For workers from coffee-producing districts, we do not find any significant impact of other price shocks on turnover. Our placebo test further strengthens the validity of our main specification.

To summarize, we leverage exogenous global shocks in coffee prices and find causal evidence that applicants would reject a job in the HIP for better outside options, but we do not find supportive evidence that such a hypothesis explains separation on the job. In this next subsection, we will use a simple job search model to rationalize these results.[8]

[7] Appendix C provides more details about how we clean the price data before analysis.

[8] In principle, our identification strategy does not inherently rely on coffee production. If turnover is driven by better outside options in the agricultural sector, any price shocks of major crops should result in similar effects on turnover. However, it is difficult to find either exogenous price shocks for all other major crops (e.g. teff, maize, wheat, all mainly for domestic use), or high-frequency price data for cash crops (e.g. chat).

3.4 Using Job Search Model to Characterize Turnover

We use a simple job search framework à la McCall (1970) to illustrate how outside options affect turnover. suppose a job seeker who already applied for a job in the HIP decides to take a job offer with wage w, or reject the offer and take an outside option with the flow income b. Suppose job seekers are from three origins (o): Urban (u), rural job seekers from non-coffee-producing districts (r), and rural job seekers from coffee-producing districts (c). The outside options in each origin follow the distribution F^o. The outside options in the rural area are characterized by the yield distribution for each crop type $h^k \sim H^k$ and the exogenously determined crop price P^k. All distributions are independent of each other.

Define the first type of turnover as T_1^o for workers with origin o, that is, the percentage of workers rejecting the offer from the HIP. Standard McCall model shows that workers' reservation wage would be equal to the flow income of their outside options. Assume that urban job seekers would only access urban distribution of outside options:

$$T_1^u = 1 - Pr(b < w) = 1 - F^u(w) \tag{2}$$

For rural workers, we assume that job seekers can access urban outside options as well as crop production, but only job seekers from coffee-producing districts can work in coffee production:

$$T_1^r = 1 - Pr(b < w \ \& \ P^k h^k < w \ \forall k) = 1 - F^u(w)F^r(w) \tag{3}$$

$$T_1^c = 1 - Pr(b < w \ \& \ P^k h^k < w \ \forall k) = 1 - F^u(w)F^r(w)H^c(\frac{w}{P^c}) \tag{4}$$

Our identification specification, Equation 1, thus captures the following:

$$\frac{\partial T_1^c}{\partial \log P^c} = F^u(w)F^r(w)h^c(\frac{w}{P^c})\frac{w}{P^c} \tag{5}$$

From the turnover data from the HIP, we can estimate the four parameters jointly: $F^u(w)$, $F^r(w)$, $H^c(w/P^c)$, and $h^c(w/P^c)$. These four parameters characterize the distribution of outside options at the point of HIP salary. In addition, We can directly measure these four distributional parameters from the survey data from ESS and AgSS; comparing the two sets of estimates would provide a test of how well the job search model fits the survey data. Table

2 shows the joint estimates of the four parameters using Generalized Methods of Moments (GMM) on Equations 2–5, as well as the survey measures using observable data from ESS and AgSS.[9] We set the HIP salary w to the median in 2022 (2,415 ETB) and coffee price P^c to the average sales price \$74 estimated from ESS.

In general, the GMM estimates of cumulative distribution F^u, F^r, and H^c are significantly different than the survey measures. This is potentially consistent with the usual critique of reservation wage method, as the measurement of reservation wage or outside options is heavily subject to noise and reservation wages often do not predict job take-up (Hall and Krueger, 2012). If anything, the model estimates of $F^u(w)$ and $H^c(w/P^c)$ are significantly higher than their corresponding survey measures, potentially implying that the HIP jobs provide good amenities that are not captured by monthly salary (e.g. career progression). The GMM estimates of h^c, however, are not distinguishable from the survey estimates. This potentially lends some credibility to the search hypothesis: The increasing turnover at the application stage in response to exogenous shocks in outside options tracks the shape of the probability density function of outside options.

Table 2: Model Estimates of Job Search Model

			Survey measures			
Parameters	Model estimates	95% CI	Assumptions	Data	Estimates	t-test
$F^u(w)$	0.739	[0.708, 0.771]	$w = 2415$	ESS	0.496	0.000
$F^r(w)$	0.779	[0.687, 0.870]	$w = 2415$	ESS	0.911	0.005
$H^c(w/P^c)$	0.933	[0.815, 1.051]	$w = 2415, P^c = 74$	ESS	0.673	0.000
				AgSS	0.452	0.000
$h^c(w/P^c)$	0.016	[0.003, 0.029]	$w = 2415, P^c = 74$	ESS	0.012	0.521
				AgSS	0.009	0.293

Notes: We estimate the four distributional parameters using Generalized Methods of Moments (GMM) on Equations 2–5 and report their 95% confidence intervals. The survey measures are obtained from Ethiopian Socio-economic Panel Survey 2021/22 (ESS) and Agricultural Sample Survey in 2018/19. We set the wage in the Hawassa Industrial Park at the median level in 2022, calculated from the personnel data. We set the price of coffee at the average sales price obtained from household surveys in ESS.

Can we use the job search framework to explain turnover after workers take up a job? We can express turnover after taking a job for workers from coffee-producing districts as

[9]For rural income distribution, we impute average yields or income per adults, potentially more comparable with the urban distribution of salary.

follows, accounting for the selection of workers at the application stage:

$$T_2^c = 1 - \frac{H^c(w/P^c)}{H^c(w/P_0^c)} \tag{6}$$

P_0^c is the coffee price at the time of application, P^c after worker takes up a job. When there is an exogenous price shock in P^c, controlling for initial coffee price P_0^c, our identification in Equation 1 would capture the following:

$$\frac{\partial T_2^c}{\partial \log P^c} = \frac{h^c(w/P^c)}{H^c(w/P_0^c)} \cdot \frac{w}{P^c} \tag{7}$$

Assuming $P^c = P_0^c$, using our estimates from Table 2, we would have expected the effect of price shocks on T_2^c to be 0.560–0.664, significantly different from zero. This confirms that search hypothesis cannot explain workers' separation from the job; otherwise, workers who selected into the HIP would have been much more responsive to the coffee price shocks.

In summary, for the 34% applicants who never took up a job assignment in the HIP, we can use search hypothesis, illustrated with the simplest job search model, to explain the extent to which their turnover decisions respond to exogenous price shocks in coffee production. For the 43% applicants who separated from the job after taking a job in the HIP, search hypothesis cannot explain the lack of turnover response to exogenous price shocks.

4 Learning Hypothesis

We now turn to a different hypothesis: Did workers quit working in the HIP because they learn about their low productivity on the job? Inspired by a classic paper by Jovanovic (1979), we first examine whether high productivity is correlated with less separation from the job, the main prediction from Jovanovic (1979). We then use a simple model to examine whether such a learning hypothesis can explain the empirical pattern.

4.1 Estimating Individual Productivity

We collected detailed personnel and productivity data from nine firms in the HIP during July–August 2025. Eight of these nine firms record productivity daily for each production

line. However, when a new worker just joined a company and was assigned to a production line, we can compare team-level productivity before and after the worker joined the production line and deduce individual-level productivity for a subset of workers at the beginning of the tenure. Specifically, we use the following specification, similar to an AKM model, to impute individual-level productivity:

$$Y_{j,i,t} = \alpha_j + \alpha_i + \alpha_t + e_{ijt} \tag{8}$$

$Y_{j,i,t}$ is the productivity for production line j where worker i worked on day t. For the most part of the analysis, we use the same productivity measure for all firms, that is, the percentage of target production met, or "efficiency". α_j are the production line fixed effects, α_i are the individual fixed effects, α_t are the day fixed effects, and e_{ijt} is the idiosyncratic error term. Assuming the production line assignment is orthogonal to individual productivity, we interpret α_i as the imputed individual productivity of worker i.

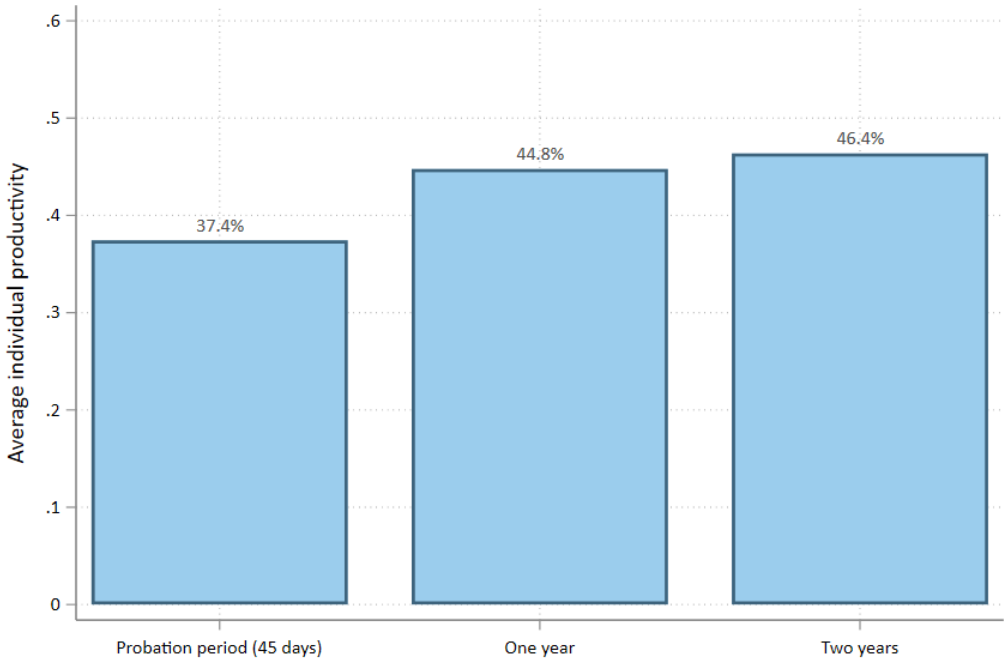
Using this method, we are able to impute individual-level productivity within probation period for 2,294 individuals, within one year for 3,371 individuals, and within two years for 3,834 individuals. Note that the imputed individual productivity within probation period is potentially more accurate because when the firm gets more information about workers' productivity, they may assign workers to different production lines based on their productivity and thus affect the causal interpretation of α_i. Figure B9 shows a large variation in the imputed individual productivity during probation.

4.2 Correlation between Turnover and Productivity

Figure 8 first shows the average imputed individual productivity within probation period, within one year, and within two years. We see an increase in individual productivity from 38% to 46% within the first year, about 20% increase, and stay at the similar level within two years. This is consistent with one of the predictions in Jovanovic (1979): Over time, as more low-productivity workers realize their real productivity and quit, the remaining workers present higher-level productivity on average. The average productivity stays similar two years after compared to one year after, suggesting that most workers may realize their productivity within one year of tenure.

Table 3 examines the correlation between individual productivity and the likelihood of separation; keep in mind that we will not observe the first type of turnover because all workers here have already accepted the job assignment. Panel A, Column 1 and 2 show that one standard deviation increase in individual productivity during probation is correlated with 3.2–4.0 percentage points decrease in the separation likelihood (standard deviation of imputed individual productivity 0.20). This is consistent with the main prediction from Jovanovic (1979), that workers may observe a negative match component on the job and separate as a result. Column 3 and 4 show that the negative correlation mainly exists in industry-specific separation, that workers separated and never returned to HIP. This suggests that these entry-level workers may learn about their comparative advantage of working in garment industry in general at the beginning of their tenure. Table B7 further breaks down turnover by separation within 60 days, 90 days, 180 days, and one year, and shows that the correlation is most significant when we select 90-day or 180-day window. This suggests that workers may make turnover decisions soon after receiving the productivity signals.

Figure 8: Average Imputed Productivity of Individual Workers



Notes: This figure shows average productivity measured within probation period (45 days), within one year, or within two years. Individual productivity is imputed using Equation 8 with personnel data collected from nine firms in the industrial park. Productivity is measured by the percentage of targets met daily.

Panel B replicates the correlations with individual productivity imputed within the first year. By definition, Panel B examines separation for workers who quit after staying in the firm for at least one year.[10] The correlation between productivity and separation remains at similar magnitudes for this selected sample of workers, albeit less precise. However, Column 3 and 4 show that separation that happens one year after joining the company is mostly firm-specific, not industry-specific. One standard deviation increase in individual productivity within one year is correlated with 4.4 pp. decrease in the likelihood of separating from the firm and returning to another firm in the HIP. The lack of correlation between productivity and industry-level separation suggests that one year after joining the firm, workers are less likely to update their belief of own comparative advantage based on a bad signal of productivity, but more likely to interpret it as a low-value firm-specific match.

In general, we find supportive correlational evidence that is consistent with the key predictions from Jovanovic (1979), that average productivity among stayers increases over time, and that separation is negatively correlated with productivity. We further find nuances regarding industry- or firm-specific separation: At the start of tenure, workers tend to update their comparative advantage for the garment industry; later into the tenure, workers are more likely to update their firm-specific match values. In the next subsection, we will attempt to use a simple learning model to rationalize our empirical findings.

4.3 Using Learning Model to Characterize Turnover

Here, we mainly discuss a simple learning model where workers receive one productivity signal at a time and infer individual-specific productivity, which guides the decision of industry-specific separation, as well as firm-specific productivity, which guides the decision of firm-specific separation. We simplify the model of Jovanovic (1979) into a two-period model, (i) start of tenure, corresponding to probation in our empirical analysis, and (ii) later into the tenure, corresponding to one year in our empirical analysis where workers' learning factors less of the productivity noise coming from the idiosyncratic match component.

Suppose at the beginning of the worker's tenure, each worker and firm both observe a

[10]The sample size of Panel B is larger because there are more workers for whom we cannot observe productivity during probation period.

Table 3: Correlation Between Productivity Measure and Turnover

Panel A. Productivity imputed during probation (45 days)

	(1)	(2)	(3)	(4)
		Separation		
VARIABLES	All	All	Not returned	Returned
Productivity during probation	-0.168	-0.200***	-0.127*	-0.0734
	(0.0895)	(0.0318)	(0.0530)	(0.0562)
	[0.157]	[0.00814]	[0.0969]	[0.283]
Data source	Personnel	HIP	HIP	HIP
Observations	2,305	1,236	1,236	1,236
R-squared	0.360	0.048	0.044	0.042
Dep var mean	0.401	0.346	0.165	0.181

Panel B. Productivity imputed within one year

	(1)	(2)	(3)	(4)
		Separation		
VARIABLES	All	All	Not returned	Returned
Productivity within one year	-0.200	-0.241	-0.0201	-0.221***
	(0.145)	(0.169)	(0.0571)	(0.0398)
	[0.263]	[0.154]	[0.742]	[0.00514]
Data source	Personnel	HIP	HIP	HIP
Observations	3,474	1,853	1,853	1,853
R-squared	0.424	0.056	0.053	0.049
Dep var mean	0.401	0.346	0.165	0.181

Notes: This table reports the correlation between individual productivity within the probation period and the likelihood of quitting the industrial park. Individual productivity is imputed using Equation 8 with personnel data collected from nine firms in the industrial park. Panel A uses productivity imputed within the probation period (45 days since joining the firm); Panel B uses productivity imputed within one year since workers joining the firm. Productivity is measured by the percentage of targets met daily. Dependent variables: Column 1 and 2— the worker separated from the job. Column 3 — the worker separated from the job and never returned to HIP. Column 3 — the worker separated from the job and returned to another firm in HIP later. Column 1 uses personnel records from individual firms; Column 2–4 uses administrative records on turnover for those who can be matched. All regressions control for firm fixed effects and cohort fixed effects (the month when the worker first joined to the firm). Standard errors are clustered two-way at the production line level and at the cohort level. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

productivity signal y_{ij}^0 for the worker generated by the following:

$$y_{ij}^0 = y_{ij} + \epsilon_{ij}^0$$

Assume the true productivity follows $y_{ij} \sim N(y_0, \sigma^2)$ and the idiosyncratic match component in each period follows $\epsilon_{ij}^0 \sim N(0, \sigma_\epsilon^2)$. Worker i would update on the true productivity y_{ij} as follows:

$$E[y_{ij}|y_{ij}^0] = y_0 + \frac{\sigma^2}{\sigma^2 + \sigma_\epsilon^2}(y_{ij}^0 - y_0) \equiv y_0 + \lambda(y_{ij}^0 - y_0)$$

Assume that when workers apply for jobs in the HIP, they all have the same belief about their productivity at y_0. To simplify the discussion of separation, assume that workers are indifferent between HIP and outside options at the application stage, that is, their productivity at outside options is also equal to y_0. We further assume an idiosyncratic switching cost for individual workers $c \sim N(0, \sigma_c^2)$ for smoothness. Worker i quits the HIP permanently if:

$$E[y_{ij}|y_{ij}^0] + c < y_0$$

Define the total separation rate at the start of the tenure T^0. To simplify notation, denote $\gamma = \lambda/\sigma_c$, $x_{ij}^0 = y_{ij}^0 - y_0$. Then we have:

$$T^0 = \Phi(-\gamma x_{ij}^0) \tag{9}$$

Now, we examine when workers observe repeated productivity signals one year into their tenure. The weight of productivity noise ϵ_{ij} would shrink by the number of signals N, $\lambda_\epsilon' = \lambda_\epsilon/N$, and workers would put most weight on their true productivity (denote $\overline{y_{ij}}$ as the average signal):

$$E[y_{ij}|\overline{y_{ij}}] = y_0 + \frac{\sigma^2}{\sigma^2 + \sigma_\epsilon^2/N}(\overline{y_{ij}} - y_0) \equiv y_0 + \lambda_\mu'(\overline{y_{ij}} - y_0)$$

Here, we need to account for the selection of worker i at the beginning of tenure because

$\overline{y_{ij}}$ is inherently correlated with the initial signal y_{ij}^0. Worker i would quit one year after if:

$$E[y_{ij}|\overline{y_{ij}}] + c < \mu_0 \text{ conditional on } E[y_{ij}|y_{ij}^0] + c > \mu_0$$

Denote $\overline{x_{ij}} = \overline{y_{ij}} - y_0$. We assume the relationship $x_{ij}^0 = A\overline{x_{ij}} + B$, which we will estimate with the sample where we can observe both productivity signals and impute for those we cannot observe the initial signal. With this assumption, the permanent separation rate one year after can be written as follows (denote $\gamma' = \lambda'/\sigma_c$):

$$T^1 = \frac{\Phi\left(-\gamma'\overline{x_{ij}}\right) - \Phi\left(-\gamma'(A\overline{x_{ij}} + B)\right)}{1 - \Phi\left(-\gamma'(A\overline{x_{ij}} + B)\right)} \tag{10}$$

Equations 9 and 10 thus match the correlational results from Table 3. We use the residual productivity after controlling for firm fixed effects and cohort fixed effects to proxy x_{ij}^0 and $\overline{x_{ij}}$; we further assume workers receive $N = 365/45 \approx 8$ signals within the one-year window. Then, we use GMM to estimate the key learning parameters. Given that we only have two moments for three underlying parameters (σ, σ_ϵ, σ_c), we set $\sigma_c = 1$ for the estimation so that our estimate directly reflects the weight that workers put on true productivity λ.

Table 4, Panel A shows the GMM results by jointly estimating Equations 9 and 10. Our estimated λ equals 0.579, that is, workers would put more than half of the weight on true productivity during the learning process. After one year, the weight λ' approaches near one. Our result is thus consistent with the insights from Jovanovic (1979), that workers learn about their productivity on the job. Such learning happens relatively fast, especially at the beginning of their tenure.

Can we extend this model to explain industry-specific and firm-specific separation? Appendix A presents an extension of the simple learning model, where workers infer two productivity components from one productivity signal in each period: Individual-specific component μ_i that guides industry-specific separation, and firm-specific component θ_j that guides firm-specific separation. $\gamma_\mu = \lambda_\mu/\sigma_c$ is the discounted weight on individual-specific component in the first period, γ'_μ is the discounted weight in the second period, likewise for γ_θ and γ'_θ.

Table 4, Panel B shows the estimates of the extended model in the first period. The estimate for γ_μ is negative, while the learning model specifies that γ_μ should be non-negative. This suggests that the learning model is potentially misspecified for the start of the tenure.

Panel C presents the joint estimate for the second period, assuming the weight on idiosyncratic match component is zero. Both estimates are significantly positive, with γ'_μ roughly equal to γ'_θ, which suggests that if the learning hypothesis holds, upon observing productivity signals for one year, workers assign roughly equal weights to individual-specific and firm-specific productivity. Panel C imposes the ratio $\gamma'_\mu/\gamma'_\theta$ on the first two parameters γ_μ and γ_θ and re-estimate Equations 15 and 16. Both estimates are not significantly different from zero; if we take the estimated values as given, when workers observe their productivity at the beginning of tenure, they would almost assign all the weight to the idiosyncratic match component ϵ^0_{ij}.

Table 4: Model Estimates of Learning Model

Parameters	Model estimates	95% CI	Estimated weights
Panel A: Only one productivity component, both probation and within one year			
λ	0.579	[0.269, 0.888]	$\lambda' = 0.947$
Panel B: Two productivity components, during probation			
γ_μ	-26.2	[-37.4, -14.9]	
γ_θ	14.2	[6.12, 22.3]	
Panel C: Two productivity components, within one year, imposing $\lambda'_\epsilon = 0$			
γ'_μ	17.0	[15.6, 18.4]	$\lambda'_\mu = 0.472$
γ'_θ	19.0	[17.3, 20.6]	$\lambda'_\theta = 0.528$
Panel D: Two productivity components, during probation, imposing $\gamma_\mu/\gamma_\theta = \gamma'_\mu/\gamma'_\theta$			
γ_μ	0.053	[-0.482, 0.587]	$\lambda_\mu = 0.001$
γ_θ	0.059	[-0.538, 0.655]	$\lambda_\theta = 0.002$
			$\lambda_\epsilon = 0.997$

Notes: We estimate the learning parameters using Generalized Methods of Moments (GMM) from the learning model and report their 95% confidence intervals. The data comes from the turnover records from the Hawassa Industrial Park as well as the imputed productivity collected from personnel records from nine major firms.

In summary, a simple learning model à la Jovanovic (1979) can explain the negative correlation between total separation and productivity in general, but it cannot fully explain why low-productivity workers tend to quit the HIP altogether at the beginning of their tenure while they tend stay within the HIP and switch to a different firm later in their tenure.

4.4 Discussion of Learning Hypothesis

Our correlational evidence produces interesting nuances that cannot be explained by a simple learning model where workers infer two components of productivity at the same time. One possibility is that workers observe more than one signal on the job. In a separate paper, Wu and Lauletta (2026) finds that workers have various beliefs about job amenities such as career ladder, and they learn about these amenities in the early stage of their tenure. With the existence of a second signal that mainly reflects the firm-specific component, when workers receive a productivity signal, they may place more weight on the individual-specific component, which can trigger a larger response in industry-specific separation at the beginning of workers' tenure.

Reassuringly, the extended learning model seems able to explain the correlation between separation and productivity later in the tenure. If we take these estimates seriously, workers put almost equal weight on both individual- and firm-specific components, but they mainly react by switching to a different firm in the HIP instead of quitting the HIP altogether. This is mainly due to selection: workers with higher one-year productivity are also less likely to have quit at the beginning of their tenure. Such a selection effect is more muted when workers decide to switch between firms within the HIP.

Regardless, we do not wish to use this overly simplified learning model to capture all the nuances of separation. Our empirical evidence is, in large part, consistent with the learning model that workers update their productivity at the beginning of their tenure on the job. To some extent, this insight normalizes turnover as a natural learning process, especially at the beginning of the tenure, which is potentially beneficial for workers to identify their comparative advantages and target better job opportunities in the long run. However, since firms bear the cost of training and pay a fixed salary for all entry-level workers, the learning hypothesis indicates potential inefficiency in the labor market (Pallais, 2014): Firms may hire fewer workers in equilibrium because workers may quit once they learn about their productivity on the job.

5 Conclusion

We study the causes of high turnover rates in the context of a flagship industrial park in Ethiopia. Combining data from multiple sources and using various identifications strategies, we find that applicants' rejection of a job in HIP can be explained by workers seeking better outside options, and that workers' separation from the job can be partially explained by on-the-job learning. Our findings suggest that high turnover is not necessarily a woe of the industrial park; it reflects the nature of a competitive local labor market with many young, inexperienced workers who learn about their comparative advantages on the job.

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A Extended Learning Model with Two Productivity Components

Suppose at the beginning of the worker's tenure, each worker and firm both observe a productivity signal y_{ij}^0 for the worker generated by the following:

$$y_{ij}^0 = \mu_i + \theta_j + \epsilon_{ij}^0$$

Assume workers-specific productivity follows $\mu_i \sim N(\mu_0, \sigma_\mu^2)$, firm-specific productivity follows $\theta_j \sim N(\theta_0, \sigma_\theta^2)$, and the idiosyncratic match component in each period follows $\epsilon_{ij}^0 \sim N(0, \sigma_\epsilon^2)$. Worker i would update on both μ_i and θ_j as follows:

$$E[\mu_i|y_{ij}^0] = \mu_0 + \frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\theta^2 + \sigma_\epsilon^2}(y_{ij}^0 - \mu_0 - \theta_0) \equiv \mu_0 + \lambda_\mu(y_{ij}^0 - \mu_0 - \theta_0)$$
$$E[\theta_j|y_{ij}^0] = \theta_0 + \frac{\sigma_\theta^2}{\sigma_\mu^2 + \sigma_\theta^2 + \sigma_\epsilon^2}(y_{ij}^0 - \mu_0 - \theta_0) \equiv \theta_0 + \lambda_\theta(y_{ij}^0 - \mu_0 - \theta_0)$$

Assume that when workers apply for jobs in the HIP, they all have the same belief about their productivity at μ_0 and the firm-specific productivity at θ_0. To simplify the discussion of separation, assume that workers are indifferent between HIP and outside options at the application stage, that is, their productivity at outside options is also equal to $\mu_0 + \theta_0$. We further assume an idiosyncratic switching cost for individual workers $c \sim N(0, \sigma_c^2)$ for smoothness. Worker i quits the HIP permanently if:

$$E[\mu_i|y_{ij}^0] + c < \mu_0$$

Define industry-specific separation rate at the start of the tenure T_d^0. To simplify notation, denote $\gamma_\mu = \lambda_\mu/\sigma_c$, $x_{ij}^0 = y_{ij}^0 - \mu_0 - \theta_0$. Then we have:

$$T_d^0 = \Phi(-\gamma_\mu x_{ij}^0) \tag{11}$$

Similarly, worker i quits the HIP temporarily and returns to a different firm if:

$$E[\mu_i|y_{ij}^0] + c \geq \mu_0 \text{ and } E[\theta_j|y_{ij}^0] + c < \theta_0$$

Define firm-specific separation rate T_f^0, and denote $\gamma_\theta = \lambda_\theta/\sigma_c$. We have:

$$T_f^0 = (1 - T_d^0)\Phi(-\gamma_\theta x_{ij}^0) \tag{12}$$

Now, we examine when workers observe repeated productivity signals one year into their tenure. The weight of productivity noise ϵ_{ij} would approach zero. Without loss of generality, we assume that one year after the tenure, worker i's beliefs about individual- and firm-specific productivity are formed by considering only noises coming from μ_i and θ_j:

$$E[\mu_i|\overline{y_{ij}}] = \mu_0 + \frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\theta^2}(\overline{y_{ij}} - \mu_0 - \theta_0) \equiv \mu_0 + \lambda'_\mu(\overline{y_{ij}} - \mu_0 - \theta_0)$$
$$E[\theta_j|\overline{y_{ij}}] = \theta_0 + \frac{\sigma_\theta^2}{\sigma_\mu^2 + \sigma_\theta^2}(\overline{y_{ij}} - \mu_0 - \theta_0) \equiv \theta_0 + \lambda'_\theta(\overline{y_{ij}} - \mu_0 - \theta_0)$$

Accounting for the selection of worker i at the beginning of tenure, worker i would quit the HIP permanently one year after if $E[\mu_i|\overline{y_{ij}}] + c < \mu_0$ conditional on $E[\mu_i|y_{ij}^0] + c > \mu_0$. Denote $\overline{x_{ij}} = \overline{y_{ij}} - \mu_0 - \theta_0$. For workers that we cannot observe the initial signal, we assume the relationship $x_{ij}^0 = A\overline{x_{ij}} + B$, which we will estimate with the sample where we can observe both productivity signals. With this assumption, the permanent separation rate one year after can be written as follows (denote $\gamma'_\mu = \lambda'_\mu/\sigma_c$):

$$T_d^1 = \frac{\Phi\left(-\gamma'_\mu\overline{x_{ij}}\right) - \Phi\left(-\gamma'_\mu(A\overline{x_{ij}} + B)\right)}{1 - \Phi\left(-\gamma'_\mu(A\overline{x_{ij}} + B)\right)} \tag{13}$$

Similarly, the temporary separation can be written as follows (denote $\gamma'_\theta = \lambda'_\theta/\sigma_c$):

$$T_f^1 = (1 - T_d^1)\frac{\Phi\left(-\gamma'_\theta\overline{x_{ij}}\right) - \Phi\left(-\gamma'_\theta(A\overline{x_{ij}} + B)\right)}{1 - \Phi\left(-\gamma'_\theta(A\overline{x_{ij}} + B)\right)} \tag{14}$$

Take the derivatives of Equations 11 and 12 with respect to x_{ij}^0:

$$\frac{\partial T_d^0}{\partial x_{ij}^0} = -\gamma_\mu \phi(-\gamma_\mu x_{ij}^0) \equiv \beta_\mu^0 \tag{15}$$

$$\frac{\partial T_f^0}{\partial x_{ij}^0} = -\beta_\theta^0 (1 - T_d^0) - \beta_\mu^0 \frac{T_f^0}{1 - T_d^0} \tag{16}$$

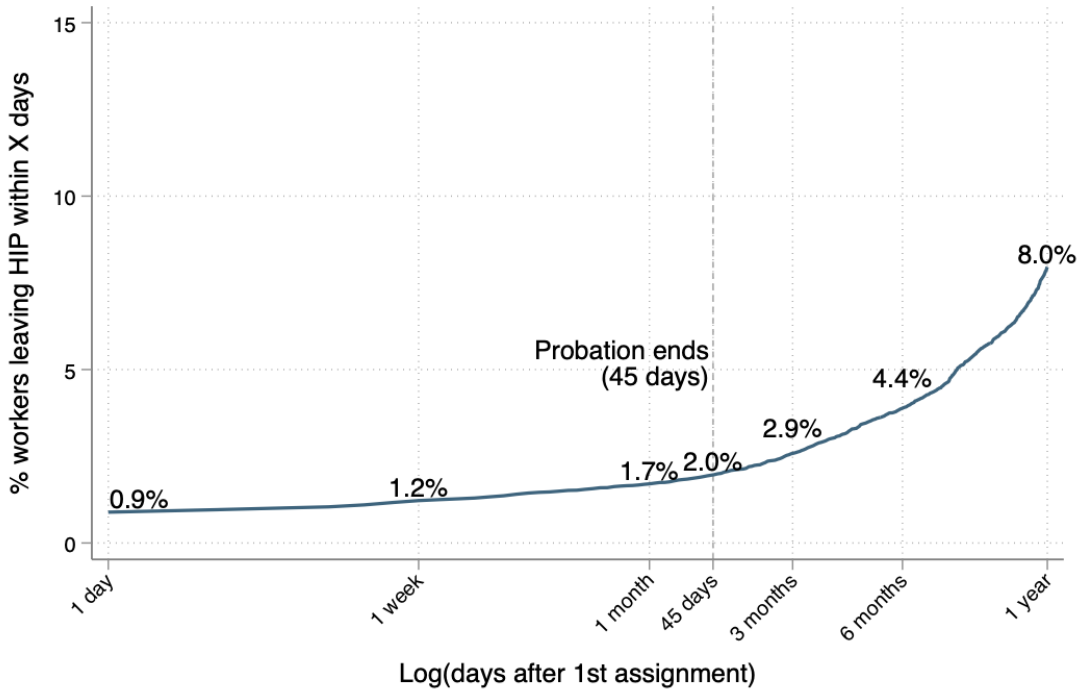
Take the derivatives of Equations 13 and 14 with respect to $\overline{x_{ij}}$:

$$\frac{\partial T_d^1}{\partial \overline{x_{ij}}} = -\gamma'_\mu \frac{\phi(-\gamma'_\mu \overline{x_{ij}})}{1 - \Phi(-\gamma'_\mu (A\overline{x_{ij}} + B))} + A\gamma'_\mu \frac{\Phi(-\gamma'_\mu \overline{x_{ij}}) - \Phi(-\gamma'_\mu (A\overline{x_{ij}} + B))}{1 - \Phi(-\gamma'_\mu (A\overline{x_{ij}} + B))} \equiv \beta_\mu^1 \tag{17}$$

$$\frac{\partial T_f^1}{\partial \overline{x_{ij}}} = -\beta_\theta^1 (1 - T_d^1) - \beta_\mu^1 \frac{T_f^1}{1 - T_d^1} \tag{18}$$

B Appendix Figures and Tables

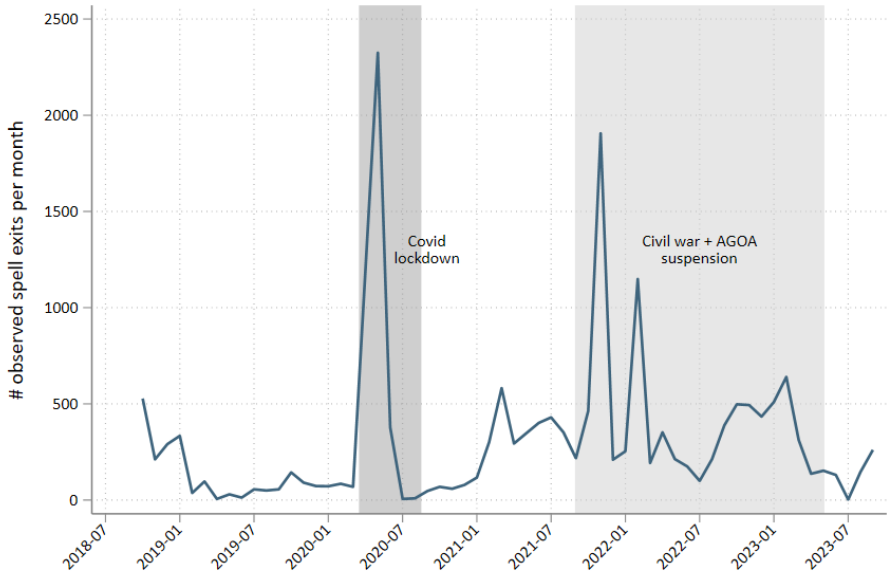
Figure B1: Percentage of Workers Separating From the First Job and Never Returning



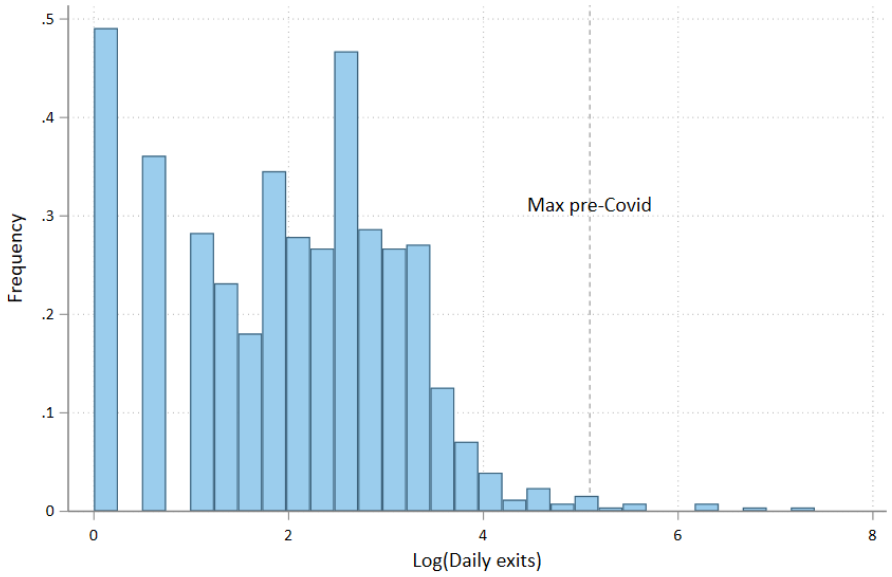
Notes: This figure shows the distribution of the turnover in the industrial park between August 2018 and September 2023. We show the key statistics of turnover within one day, one week, one month, 45 days when the probation period ended, three months, six months, and one year.

Figure B2: Mass Layoff in HIP

Panel A. Monthly Trend of Separation



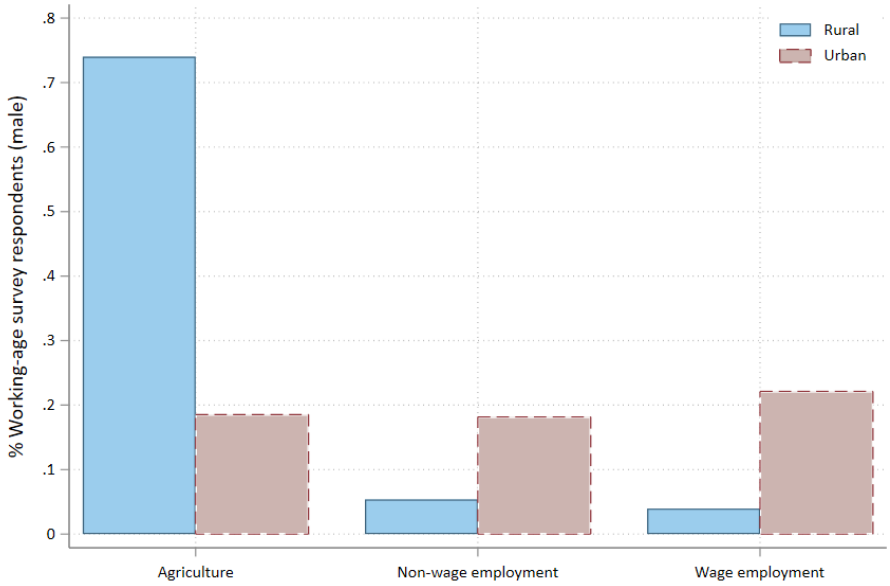
Panel B. Histogram of the Number of Daily Separations



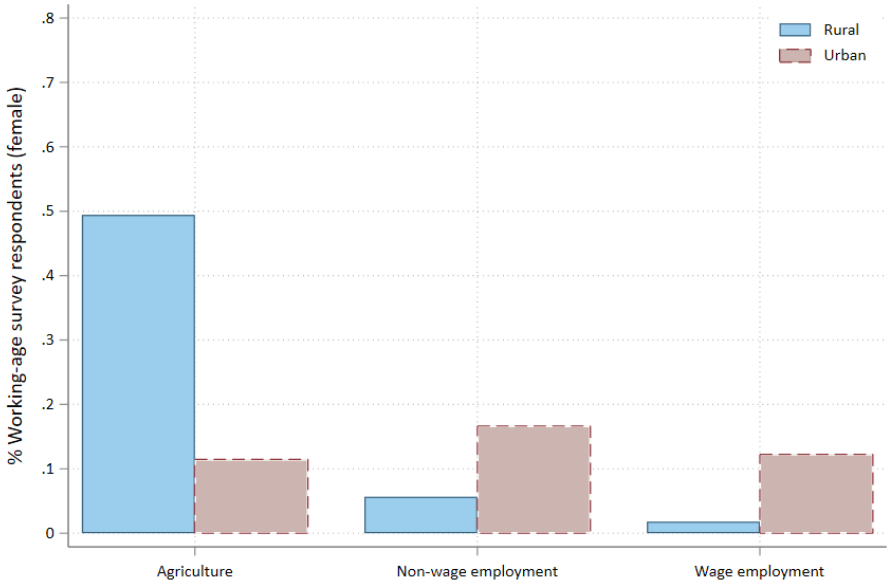
Notes: This figure shows the trend of monthly separation between August 2018 and September 2023 (Panel A) and the histogram of the number of daily separations (Panel B). Panel A indicates two major periods of mass layoffs (Covid, civil war and AGOA suspension). The gray line in Panel B indicates the maximum daily separation before Covid.

Figure B3: Distribution of Types of Employment, By Rural and Urban

Panel A. Male workers



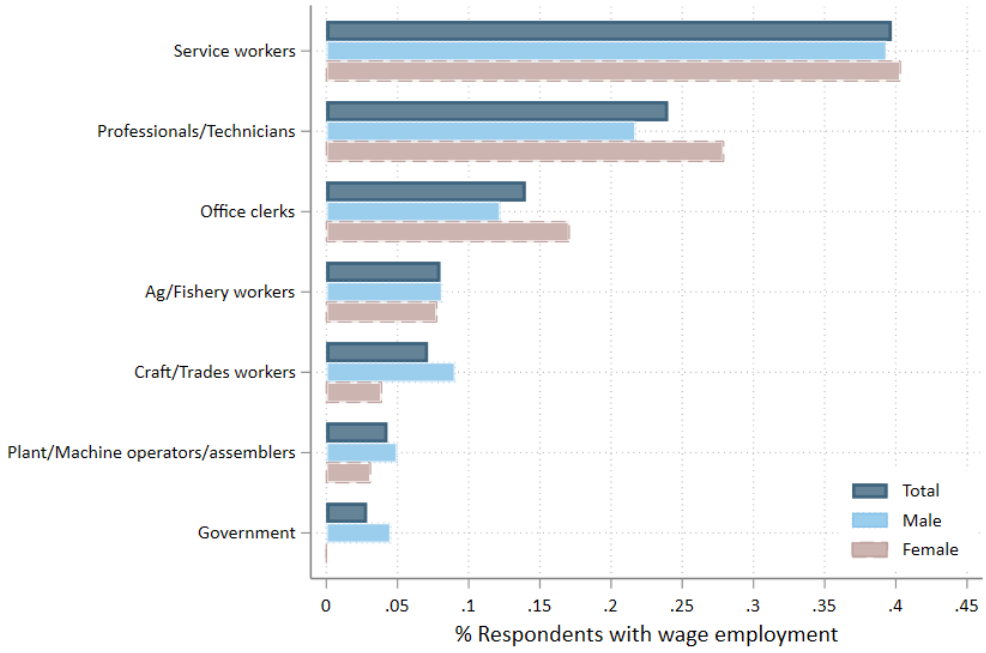
Panel B. Female workers



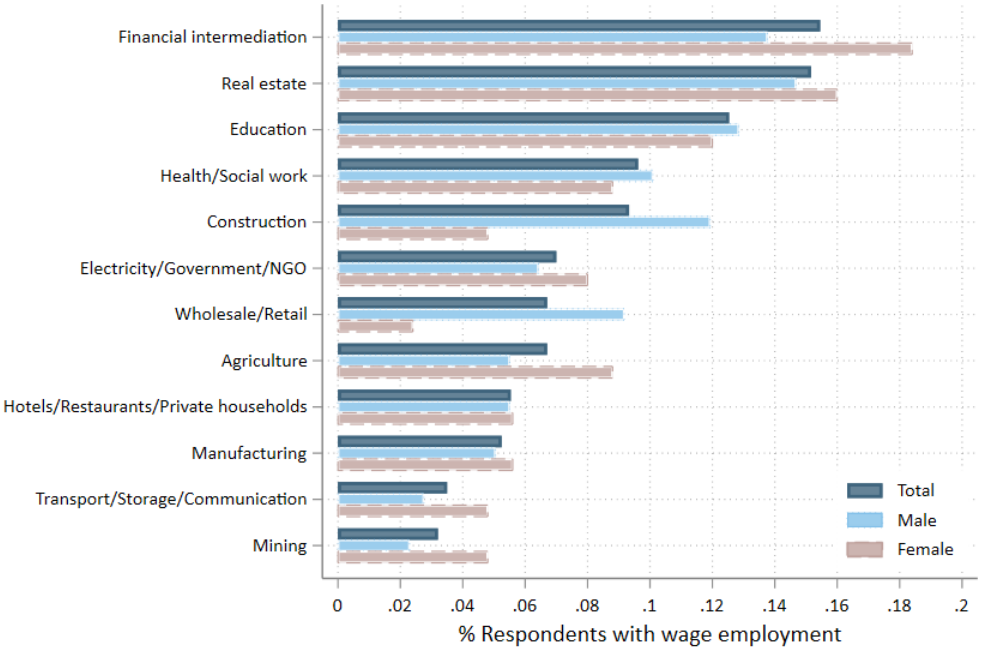
Notes: This figure shows the distribution of types of work in SNNP and Oromia from LSMS data. Only individuals aged above 15 and below 64 are included.

Figure B4: Descriptives of Wage Employment in SNNPR and Oromia Regions

Panel A. Occupation Distribution

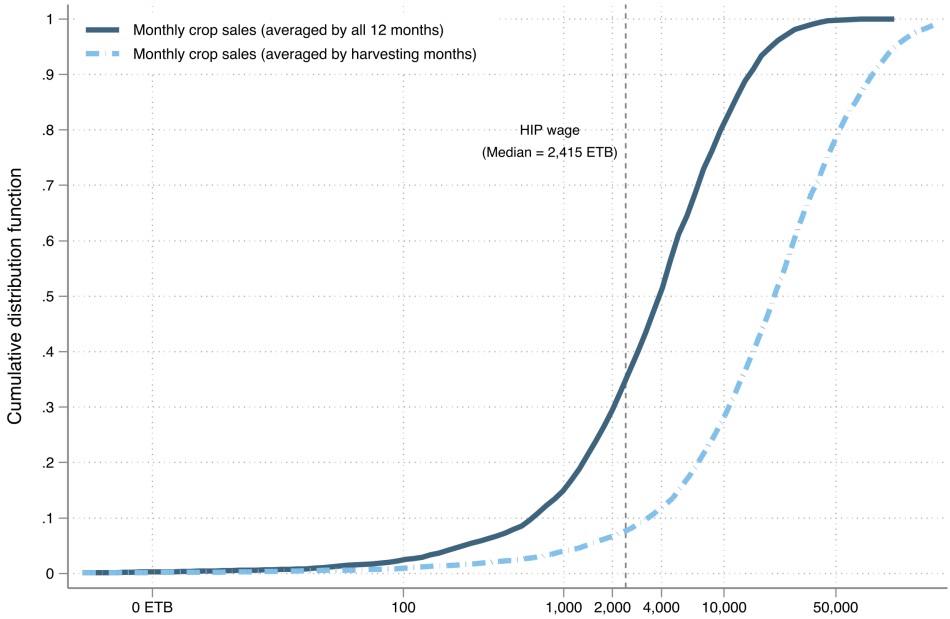


Panel B. Industry Distribution



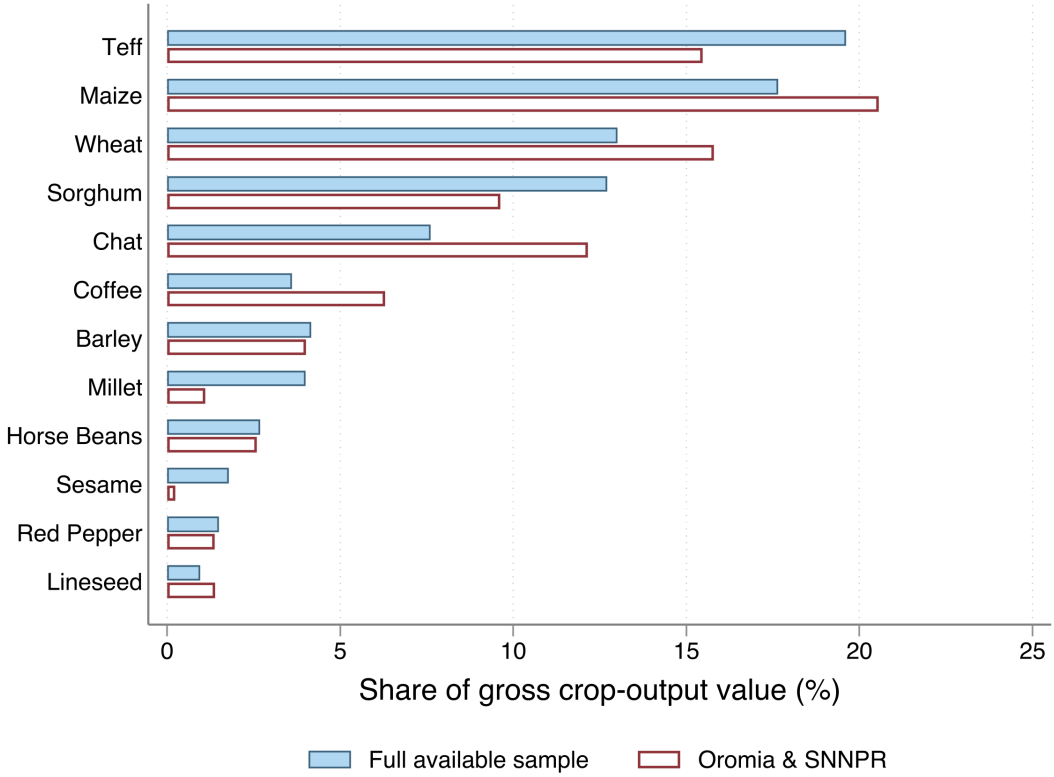
Notes: We use Ethiopian Socio-Economic Panel Survey 2021–22 to describe the distribution of occupations among workers with wage employment in SNNPR and Oromia regions. Only individuals aged above 15 and below 64 are included.

Figure B5: Distribution of Imputed Monthly Income in Agricultural Sectors



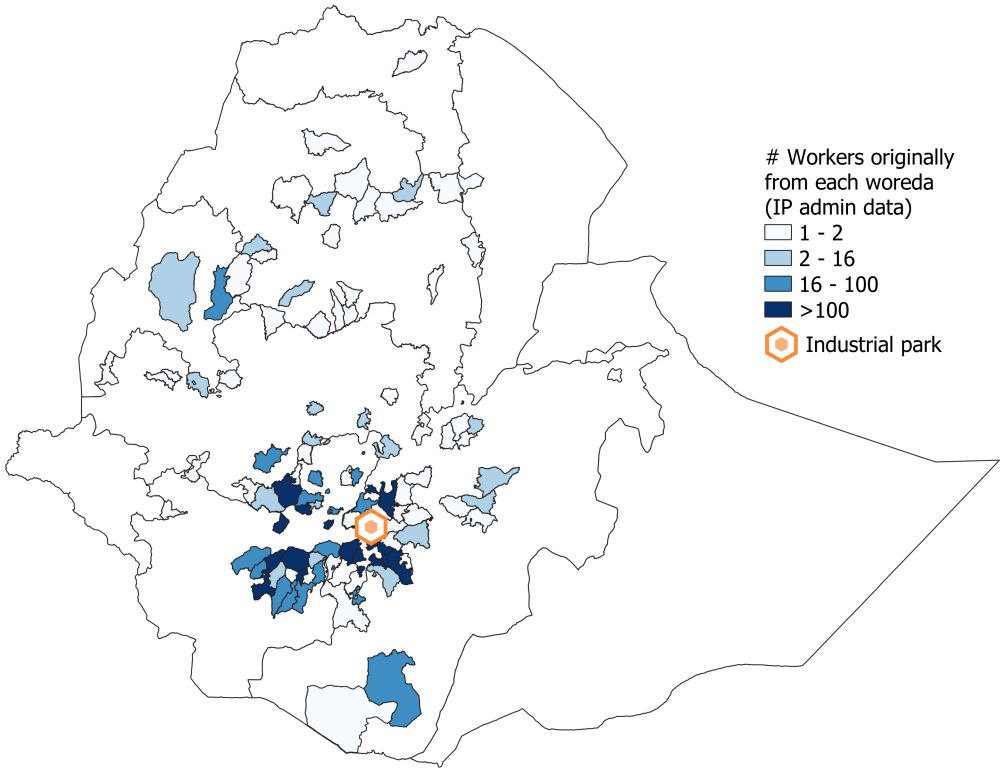
Notes: We use 2018/19 Ethiopian Agricultural Sample Survey (AgSS) for the crop-field and household data and Ethiopian Socio-Economic Panel Survey 2021–22 for the price data. Then we describe the distribution of imputed agricultural income in SNNPR and Oromia regions. For each crop production, we multiply the total quantity harvested in the last 12 months for each field with the average price across all households in the sample. The blue solid line plots the CDF of imputed income divided by 12. The light blue, dashed line plots the CDF of imputed income divided by the average number of harvesting months for each crop.

Figure B6: Common Crop Production by Imputed Gross Output Value



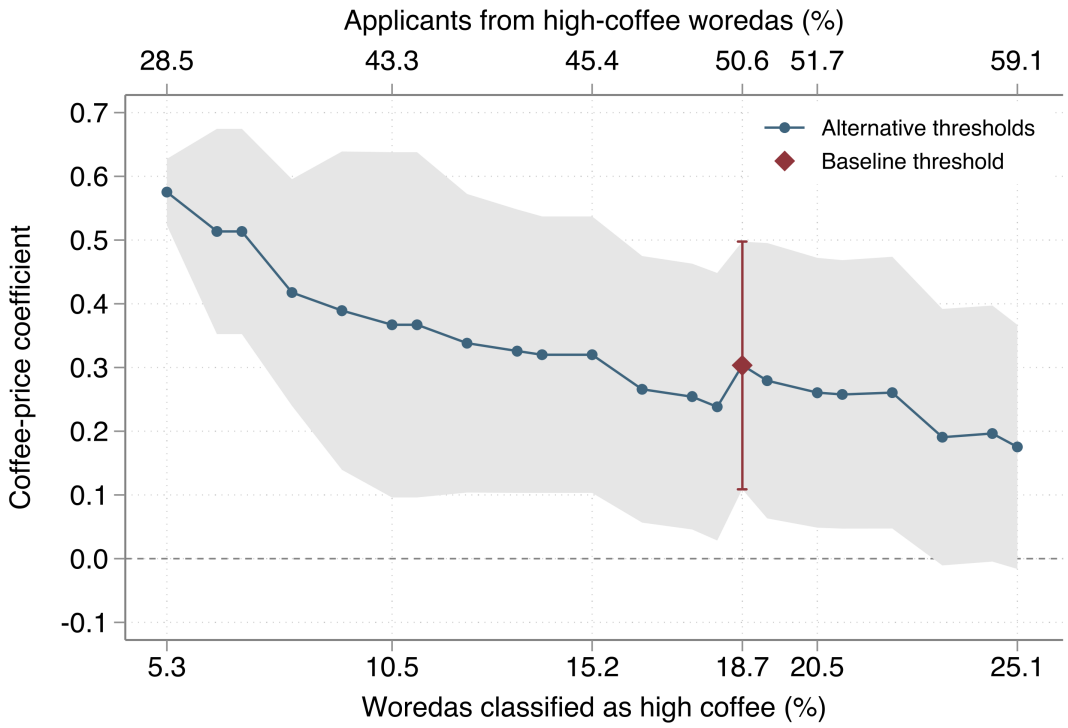
Notes: This figure shows the share of imputed, aggregate value of gross crop outputs for each crop, using crop production data and household sampling weights from the 2018/19 Ethiopian Agricultural Sample Survey (AgSS) and the price data from Ethiopian Socio-Economic Panel Survey (ESS) 2021–22. Blue bars represent the full available sample; red outlined bars represent the Oromia and SNNPR subsample. Output is valued by multiplying crop-specific ESS 2021–22 prices, except for coffee, which uses the ECX local unwashed coffee price. Households with any unpriced crop, or incomplete production quantities are excluded. Output value includes production consumed within the household and is measured before deducting production costs.

Figure B7: Distribution of Workers' Origins



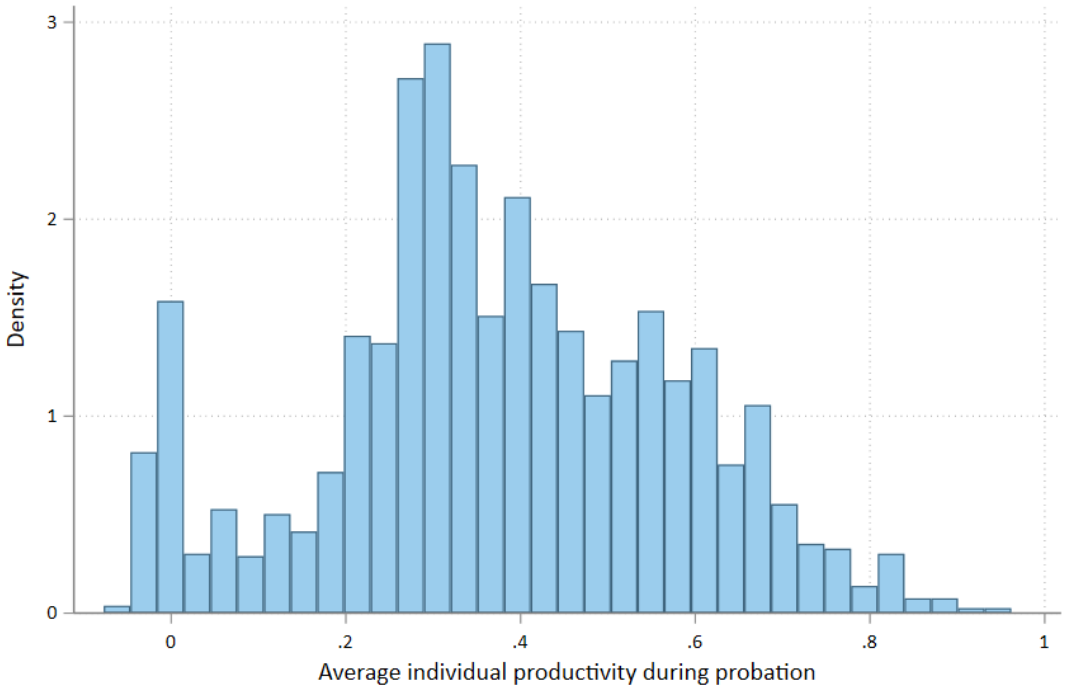
Notes: This figure shows the distribution of workers' origins. We only show the districts from which at least one worker in the administrative data was originated.

Figure B8: Sensitivity of High Coffee Exposure Threshold



Notes: This figure reports the coefficient on high-coffee exposure interacted with the log global coffee price one month after application in the non-entry regression. Alternative thresholds classify the top 5% through top 25% of 171 matched origin woredas as high exposure, in one-percentage-point steps. Woredas are ranked by their unweighted coffee crop-record share in the 2018/19 Agricultural Sample Survey, retaining ties. The lower axis shows the actual percentage of woredas classified as high exposure; the upper axis shows the corresponding percentage of the 14,873 applicants in the estimation sample. Paired labels refer to identical cutoffs. The red diamond marks the baseline applicant-median cutoff of 11.52% coffee exposure, covering 18.7% of woredas and 50.6% of applicants. Every regression controls for the same exposure indicator interacted with log coffee price one month before application, gender, education, age, woreda fixed effects, and application year-month fixed effects. Shading and the baseline error bar show pointwise 95% confidence intervals using two-way clustering by woreda and application year-month.

Figure B9: Productivity Measure Histogram



Notes: This figure shows the histogram of individual productivity imputed within probation period (45 days). Individual productivity is imputed using Equation 8 with personnel data collected from nine firms in the industrial park. Productivity is measured by the percentage of targets met daily.

Table B1: Predicting Local Coffee Prices Using Global Coffee Prices

	(1)	(2)	(3)	(4)
	Log price: Local coffee			
	Washed	Washed	Unwashed	Unwashed
Log price: Global, Robustas	0.984***		1.049***	
	(0.121)		(0.145)	
Log price: Global, Other Mild Arabica		0.816***		0.932***
		(0.0452)		(0.0453)
Observations	53	53	53	53
R-squared	0.478	0.830	0.425	0.846
P-value: coefficient = 1	0.894	0.000	0.738	0.138

Notes: This table reports the correlation between local coffee prices and global coffee prices. The local coffee prices are obtained from Ethiopia Commodity Exchange. The global coffee prices are obtained from International Monetary Fund. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Table B2: Causal Effect of Coffee Price Shocks on Turnover: IV Specification

	(1)	(2)	(3)	(4)
	Dependent variable: Never worked in HIP			
$C_j \times \log P_{t+1}$, Local Washed	0.112		0.500***	0.153**
	(0.101)		(0.182)	(0.0716)
	[0.272]		[0.00807]	[0.0374]
$C_j \times \log P_{t+1}$, Local Unwashed		0.00947		
		(0.133)		
		[0.944]		
Observations	14,873	14,873	14,873	14,873
R-squared	0.404	0.403	0.027	0.028
Control mean	0.424	0.424	0.424	0.424
F-stat			10.87	18.57

Notes: C_j is the fixed high-coffee exposure indicator defined in Table 1, using the applicant-sample median of unweighted crop-record shares. Urban Hawassa city is excluded; rural Hawassa/Awasa Zuria remains included. Month t is initial registration. Dependent variable is whether a worker never took up a job in the HIP and left no hiring records in the HIP administrative data. Column 1 uses local washed coffee prices, Column 2 local unwashed prices, Column 3 instruments local washed prices with global Arabica prices, and Column 4 instruments local washed prices with global Robusta prices. The pre-price interaction uses global Arabica at $t-1$ in Columns 1–3 and global Robusta in Column 4. Local monthly prices are built from Ethiopia Commodity Exchange daily prices; eight missing months are linearly interpolated. Global prices are from the International Monetary Fund. All columns include demographics, origin and registration year-month fixed effects, with two-way clustering by origin and year-month. Control means refer to low-coffee origins in each estimation sample. The F-stat row reports the Kleibergen–Paap rk Wald F statistic. Standard errors in parentheses; p-values in brackets. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Table B3: Causal Effect of Coffee Price Shocks on Turnover by Seasonality

	(1)	(2)	(3)	(4)	(5)
	Never worked	Separated			
	in HIP	All	Not returned	Returned	All
$C_j \times \log P_{t+1} \times$ harvest	0.398***	0.0817	0.0226	0.0591	-0.108
	(0.116)	(0.122)	(0.0910)	(0.0863)	(0.233)
	[0.00117]	[0.506]	[0.805]	[0.496]	[0.641]
$C_j \times \log P_{t+1} \times$ lean	0.404***	0.0867	0.0259	0.0608	-0.0941
	(0.118)	(0.122)	(0.0887)	(0.0877)	(0.231)
	[0.00121]	[0.479]	[0.771]	[0.491]	[0.684]
Observations	14,873	9,533	9,533	9,533	2,081
R-squared	0.404	0.138	0.148	0.075	0.691
Control mean	0.424	0.201	0.0816	0.119	0.201

Notes: $C_j = 1$ for origins at or above the applicant-sample median coffee exposure. Coffee exposure is the unweighted share of fields reporting coffee in Agricultural Sample Survey; the median is 10% among districts where at least one applicant in HIP was originated. Urban Sidama/Hawassa city is excluded; rural Hawassa/Awasa Zuria remains included. Month t is initial date of registration in Column 1 and the first day of work in Columns 2–5. P_{t+1} is the global price of other mild Arabica coffee at $t+1$. We further interact the main independent variable with whether $t+1$ falls in the October–December harvest season. Dependent variable: Column 1, whether a worker never took up a job in the HIP and left no hiring records in the HIP administrative data. Column 2, whether a worker quit their first job. Column 3, whether a worker quit their first job and never returned to the HIP within the observational window. Column 4, whether a worker quit their first job and returned to the HIP within the observational window. Column 5, whether a worker quit their current job. All columns include gender, education, age, year-month fixed effects, origin fixed effects, two-way clustering by origin and year-month; we further control for $C_j \times \log P_{t-1}$ to control for potential selection induced by coffee prices at $t-1$. Column 5 replaces origin fixed effects with worker fixed effects and clusters at the worker level. Low-coffee reference means use the pooled eligible applicant/episode samples before regression exclusions, including in Column 5. Standard errors in parentheses; p-values in brackets. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Table B4: Causal Effect of Coffee Price Shocks in Later Months on Turnover

	(1)	(2)	(3)	(4)	(5)
	Never worked	Separated			
	in HIP	All	Not returned	Returned	All
$C_j \times \log P_{[t+2,t+12]}$	0.147	-0.00987	0.00368	-0.0136	-0.0392
	(0.0976)	(0.0845)	(0.0727)	(0.0665)	(0.133)
	[0.139]	[0.907]	[0.960]	[0.839]	[0.768]
Observations	14,873	9,533	9,533	9,533	2,081
R-squared	0.404	0.137	0.148	0.075	0.690
Control mean	0.424	0.202	0.0817	0.120	0.581

Notes: C_j is the fixed high-coffee exposure indicator defined in Table 1. Urban Hawassa city is excluded; rural Hawassa/Awasa Zuria remains included. Month t is initial registration for non-entry and episode start for separation. Never worked in HIP means no recorded allocation by the cutoff; return status uses the observed window. $P_{[t+2,t+12]}$ is the log arithmetic mean of available Arabica prices in $t+2$–$t+12$; a complete 11-month window is not required. Dependent variable: Column 1, whether a worker never took up a job in the HIP and left no hiring records in the HIP administrative data. Column 2, whether a worker quit their first job. Column 3, whether a worker quit their first job and never returned to the HIP within the observational window. Column 4, whether a worker quit their first job and returned to the HIP within the observational window. Column 5, whether a worker quit their current job. All columns include gender, education, age, year-month fixed effects, origin fixed effects, two-way clustering by origin and year-month; we further control for $C_j \times \log P_{t-1}$ to control for potential selection induced by coffee prices at $t-1$. Column 5 replaces origin fixed effects with worker fixed effects and clusters at the worker level. Low-coffee reference means use the pooled eligible applicant/episode samples before regression exclusions, including in Column 5. Standard errors in parentheses; p-values in brackets. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Table B5: Specification Tests for the Causal Effect of Coffee Price Shocks on Turnover

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent variable: Never worked in HIP							
$C_j \times \log P_{t+1}$	0.303***	0.332**	0.319**				0.282*	0.330**
	(0.0968)	(0.135)	(0.134)				(0.140)	(0.130)
	[0.00281]	[0.0137]	[0.0173]				[0.0509]	[0.0142]
$C_j \times P_{t+1}$				0.00165**				
				(0.000658)				
				[0.0150]				
$C_j \times 1(P_{t+1} > P_{t-1})$					0.0571***			
					(0.0169)			
					[0.00138]			
$C_j \times \log P_{t+1}$, Robusta						0.337**		
						(0.132)		
						[0.0133]		
Observations	14,873	13,929	13,929	14,873	14,873	14,873	10,910	14,752
R-squared	0.404			0.404	0.404	0.404	0.382	0.437
Control mean	0.424	0.468	0.468	0.424	0.424	0.424	0.492	0.422
Excluding mass-layoff episodes	No	No	No	No	No	No	Yes	No
Woreda × calendar-month FE	No	No	No	No	No	No	No	Yes

Notes: C_j is the fixed high-coffee exposure indicator defined in Table 1. Urban Hawassa city is excluded; rural Hawassa/Awasa Zuria remains included. Month t is initial registration for non-entry and episode start for separation. Dependent variable is whether a worker never took up a job in the HIP and left no hiring records in the HIP administrative data. Columns 1–7 report the main specification (1), logit (2) and probit (3) average marginal effects, price levels (4), a positive-price-change indicator (5), Robusta prices (6), and a high-daily-exit cohort exclusion (7): $t+1$ contains no day with eligible recorded exits above p95. Column 8 adds woreda × calendar-month fixed effects, allowing each origin its own seasonal pattern across the 12 months, repeated across years; 121 additional observations are excluded as singletons. All models include demographics, origin and year-month fixed effects. Pre-price controls use Arabica log prices at $t-1$, except Column 4 uses levels and Column 6 uses Robusta prices. Linear models cluster by origin and year-month; logit/probit cluster by origin. Control means use low-coffee origins in each estimation sample. Standard errors in parentheses; p-values in brackets. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Table B6: Other-Crop Price Comparisons: Coffee Exposure

	(1)	(2)	(3)	(4)	(5)
	Never worked	Separated			
	in HIP	All	Not returned	Returned	All
$C_j \times \log P_{t+1}^{\text{Maize}}$	-0.136	0.135	0.0495	0.0856	-0.183
	(0.250)	(0.139)	(0.122)	(0.0587)	(0.253)
	[0.588]	[0.335]	[0.686]	[0.150]	[0.470]
$C_j \times \log P_{t+1}^{\text{Teff}}$	-0.0944	-0.0360	-0.0435	0.00752	-0.0193
	(0.288)	(0.125)	(0.0853)	(0.0909)	(0.241)
	[0.744]	[0.774]	[0.612]	[0.934]	[0.936]
$C_j \times \log P_{t+1}^{\text{Wheat}}$	-0.121	0.0366	0.0677	-0.0311†	0.246
	(0.180)	(0.0944)	(0.0684)	(0.0777)	(0.242)
	[0.504]	[0.699]	[0.327]	[0.690]	[0.310]
$C_j \times \log P_{t+1}^{\text{Sorghum}}$	-0.213	0.0960	0.0687	0.0273	-0.500*
	(0.177)	(0.183)	(0.152)	(0.0804)	(0.281)
	[0.235]	[0.603]	[0.653]	[0.736]	[0.0760]
Coffee-benchmark observations	14,873	9,533	9,533	9,533	2,081
Coffee reference mean	0.424	0.201	0.0816	0.119	0.201

Notes: This table keeps the coffee-district indicator and replaces the coffee price with each other crop's national monthly price, using the Table 1 specification. Cereal prices are national monthly log price indices estimated from World Food Programme market-monitor quotations (HDX): all retail and wholesale quotations per 100 kg are pooled with a fixed effect for each market-variety-price-type series, and the year-month effects form the index (markets per month in 2018–2023: maize 35, teff 15, wheat 30, sorghum 20). Month t is initial registration in Column 1 and episode start in Columns 2–5.Dependent variable: Column 1, whether a worker never took up a job in the HIP and left no hiring records in the HIP administrative data. Column 2, whether a worker quit their first job. Column 3, whether a worker quit their first job and never returned to the HIP within the observational window. Column 4, whether a worker quit their first job and returned to the HIP within the observational window. Column 5, whether a worker quit their current job. All columns include gender, education, age, year-month fixed effects, origin fixed effects, two-way clustering by origin and year-month; we further control for $C_j \times \log P_{t-1}$ to control for potential selection induced by coffee prices at $t-1$. Column 5 replaces origin fixed effects with worker fixed effects and clusters at the worker level. Bottom rows give common baseline coffee-benchmark observations and low-coffee reference means, not crop-specific sample summaries; the means use the estimation sample in Column 1 and eligible episodes before regression exclusions in Columns 2–5. † marks origin-clustered inference used when the two-way clustered post-price standard error was missing or zero. Standard errors in parentheses; p-values in brackets. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Table B7: Correlation Between Productivity Measure and Turnover by Different Windows

Panel A. All separation

	(1)	(2)	(3)	(4)
	Separation, All			
VARIABLES	w/n 60 days	w/n 90 days	w/n 180 days	w/n one year
Productivity during probation	-0.0549	-0.0838**	-0.174	-0.223
	(0.148)	(0.0243)	(0.131)	(0.154)
	[0.712]	[0.0410]	[0.186]	[0.149]
Observations	1,236	1,236	1,236	1,236
R-squared	0.085	0.080	0.082	0.066
Control mean	0.203	0.213	0.231	0.259

Panel B. Industry-specific separation

	(1)	(2)	(3)	(4)
	Separation and Never Returning to HIP			
VARIABLES	w/n 60 days	w/n 90 days	w/n 180 days	w/n one year
Productivity during probation	-0.0250	-0.0428**	-0.106**	-0.134
	(0.160)	(0.0110)	(0.0203)	(0.314)
	[0.876]	[0.0298]	[0.0136]	[0.670]
Observations	1,236	1,236	1,236	1,236
R-squared	0.056	0.050	0.050	0.043
Control mean	0.0824	0.0885	0.0971	0.115

Panel C. Firm-specific separation

	(1)	(2)	(3)	(4)
	Separation and Returning to Another Firm			
VARIABLES	w/n 60 days	w/n 90 days	w/n 180 days	w/n one year
Productivity during probation	-0.0299	-0.0410	-0.0676	-0.0887
	(0.231)	(0.0661)	(0.0568)	(0.197)
	[0.897]	[0.579]	[0.319]	[0.653]
Observations	1,236	1,236	1,236	1,236
R-squared	0.052	0.049	0.052	0.043
Control mean	0.120	0.124	0.134	0.144

Notes: This table reports the correlation between individual productivity within the probation period and the likelihood of quitting the industrial park within 60 days, 90 days, 180 days, or one year. Individual productivity is imputed using Equation 8 with personnel data collected from nine firms in the industrial park. Panel A examines all separation; Panel B examines industry-specific separation where workers never return to the HIP; Panel C examines firm-specific separation where workers return to another firm in the HIP. Productivity is measured by the percentage of targets met daily. All regressions use administrative records on turnover for those who can be matched. All regressions control for firm fixed effects and cohort fixed effects (the month when the worker first joined to the firm). Standard errors are clustered two-way at the production line level and at the cohort level. * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

C Price construction

Domestic coffee prices. We construct separate monthly series for Local Washed and Local Unwashed coffee using ECX quotations and ECX daily averages republished by 2merkato. For dates with contract-level records, we calculate a trading-volume-weighted average of closing prices within each category. When both sources cover the same date and category, we retain the republished daily average. Quotations in Ethiopian Birr per Feresulla are divided by 17 to obtain Birr per kilogram. We then calculate the arithmetic mean of the available daily prices within each month, giving each observed date equal weight. Eight months lack observations during 2018–2023; these are linearly interpolated in monthly price levels before taking logarithms. The resulting series measure nominal domestic exchange prices and are common across workers' origin districts.

Domestic cereal-price indices. We use World Food Programme market-monitoring data for Ethiopia, distributed through the Humanitarian Data Exchange (HDX), to construct separate monthly indices for maize, teff, wheat, and sorghum. We retain quotations flagged as actual observations, denominated in Ethiopian Birr per 100 kilograms, and classified as retail or wholesale. A price series is defined by the combination of market, commodity variety, and price type. Multiple quotations within a series and month are averaged in logarithms, and series observed in fewer than two months are excluded. For each crop k, we estimate by OLS:

$$\log q_{st}^k = \alpha_s^k + \tau_t^k + u_{st}^k,$$

where q_{st}^k denotes the quoted price for series s in year-month t. Series fixed effects absorb persistent price-level differences across markets, varieties, and retail versus wholesale transactions. Year-month fixed effects capture the common monthly price movement. Each observed series-month receives equal weight. We pool retail and wholesale series because their coverage differs over time. Overlapping observations connect the series, under the maintained assumption that their relative price-level differences are constant over time. We construct the log-price index as

$$\log P_t^k = \widehat{\tau}_t^k - \frac{1}{12} \sum_{r \in 2018} \widehat{\tau}_r^k.$$

Thus, the log index has a zero mean in 2018. Months without quotations between observed months are linearly interpolated in the log index; we do not extrapolate beyond observed coverage. During 2018–2023, interpolation is required for June 2019 for all four crops and additionally December 2019 for teff. These indices measure domestic price movements rather than absolute Birr-per-kilogram price levels.

Alternative cereal prices. For the robustness specifications, we use the World Bank's Real-Time Food Prices data. For each crop and month, we take the arithmetic mean of the available market-level closing-price estimates and then take its logarithm. These series incorporate model-based estimates for missing observations and are used without additional interpolation. Because the World Bank series draws partly on WFP data, it provides an alternative price construction rather than an independent set of observed quotations.